MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, ADJUSTABLE, POSITIVE, VOLTAGE REGULATORS, MONOLITHIC SILICON

This specificatior is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for three- and four-terminal, monolithic silicon, adjustable, positive, voltage regulators. Three product assurance classes and a choice of lead finish are provided for each type and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit	Case outline letter
01	4-terminal voltage regulator, 5 volts $\leq V_O \leq$ 30 volts at 0.5 A	X
02	4-terminal voltage regulator, 5 volts $\leq V_O \leq$ 30 volts at 1.5 A	Y
03	3-terminal voltage regulator, 1.25 volts $\leq V_O \leq$ 37 volts at 0.5 A	X
04	3-terminal voltage regulator, 1.25 volts $\leq V_O \leq$ 37 volts at 1.5 A	Y
05	3-terminal voltage regulator, 1.25 volts $\leq V_O \leq$ 37 volts at 3.0 A	Y
06	3-terminal voltage regulator, 1.25 $\leq V_O \leq$ 37 volts at 5.0 A	Y

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline
X (device type 01)	See figure 1
Y (device type 02)	See figure 2
X (device type 03)	See figure 3
Y (device type 04,05,and 06)	See figure 4

1.3 Absolute maximum ratings.

Input voltage (device types 01 and 02)	40 Vdc
Input-output differential voltage (device types 03 and 04)	40 Vdc
(device types 05 and 06)	35 Vdc
Lead temperature (soldering, 60 seconds)	300°C
Junction temperature	T_J = 150°C 1/
Storage temperature	-65°C to +150°C

1/ The device is protected by a thermal shutdown circuit which is designed to turn off the output transistor whenever the device junction temperature is in excess of 150°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center RADC (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Input voltage range:
Device types 01 and 02 - - - - - - - - - - - - - - - 8 to 38 Vdc
Device types 03 and 04 - - - - - - - - - - - - - - - 4.25 to 41.25 Vdc
Device types 05 and 06 - - - - - - - - - - - - - - - 4.25 to 36.25 Vdc
Case operating temperature range- - - - - - - - - - -55°C to +125°C

1.5 Power and thermal characteristics.

$T_C=T_S$	Case	Max θ_{J-A}	Maximum P_D without heat sink	Max θ_{J-C}	Maximum P_D with heat sink	Max θ_{C-S} 2/
125°C 3/	X	140°C/W	0.18 W	40°C/W	0.5 W	10°C/W
	Y	35°C/W	0.71 W	4°C/W 4/	5.6 W 5/	0.5°C/W
25°C 3/	X	140°C/W	0.89 W	40°C/W	2.50 W	10°C/W
	Y	35°C/W	3.60 W	4°C/W 4/	28.00 W 6/	0.5°C/W
-55°C 3/	X	140°C/W	1.50 W	40°C/W	4.00 W	10°C/W
	Y	35°C/W	5.80 W	4°C/W 4/	45.00 W 6/	0.5°C/W

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS.

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

2/ This value represents the maximum allowable thermal impedance of a heat sink to remain within the above thermal ratings.

3/ Based on T_J = 150°C and specified values of θ_{J-A} and θ_{J-C}.

4/ Maximum θ_{JC} at all temperatures (for case Y only) = 1.5°C/W for device type 05;
= 1.0°C/W for device type 06.

5/ P_D at 125°C (for case Y only) = 12.5 W for device type 05;
= 16.6 W for device type 06.

6/ P_D at -55°C and 25°C (for case Y only) = 30 W for device type 05;
= 50 W for device type 06.

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TABLE I. Electrical performance characteristics for device type 01 (see 3.4 unless otherwise specified). 1/ 2/

Characteristic	Symbol	Condition: (fig. 12 unless otherwise stated)			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Output voltage	V_{OUT}	V_{IN} = 8V	I_L = -5mA, -500mA		4.75	5.25	V
		V_{IN} = 30V	I_L = -5mA, -50mA		4.75	5.25	V
		V_{IN} = 38V	I_L = -500mA		28.5	31.5	V
		V_{IN} = 10V	I_L = -5mA	T_C = 150°C	4.75	5.25	V
Line regulation	V_{RLINE}	8V ≤ V_{IN} ≤ 30V	I_L = -50mA		-150	150	mV
		8V ≤ V_{IN} ≤ 25V	I_L = -350mA		-50	50	mV
Load regulation	V_{RLOAD}	V_{IN} = 10V	-500mA ≤ I_L ≤ -5mA		-100	100	mV
		V_{IN} = 30V	-50mA ≤ I_L ≤ -5mA		-150	150	mV
Thermal regulation	V_{RTH}	V_{IN} = 15V	I_L = -500mA	T_C = 25°C	-50	50	mV
Standby current drain	I_{SCD}	V_{IN} = 10V	I_L = -5mA		-7.0	-0.5	mA
		V_{IN} = 30V	I_L = -5mA		-8.0	-0.5	mA
Standby current drain change versus line voltage	ΔI_{SCD} (LINE)	8V ≤ V_{IN} ≤ 30V	I_L = -5mA		-1.0	1.0	mA
Standby current drain change versus load current	ΔI_{SCD} (LOAD)	V_{IN} = 10V	-500mA ≤ I_L ≤ -5mA		-0.5	0.5	mA
Control pin current	I_{CTL}	V_{IN} = 10V	I_L = -350mA	T_C = 25°C	-5.0	-0.01	µA
				-55°C ≤ T_C ≤ 125°C	-8.0	-0.01	µA
Output short circuit current	I_{OS1}	V_{IN} = 10V			-2.0	-0.50	A
	I_{OS2}	V_{IN} = 30V			-1.0	-0.01	A
Output voltage recovery after output short circuit current	V_{OUT} (RECOV)	V_{IN} = 10V	R_L = 10Ω, C_L = 20 µF	after I_{OS1} 3/	4.75	5.25	V
		V_{IN} = 30V	R_L = 1 kΩ	after I_{OS2} 3/	4.75	5.25	V
Voltage start-up	V_{START}	V_{IN} = 8V	R_L = 10Ω, C_L = 20µF		4.75	5.25	V
Ripple rejection	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	V_{IN} = 10V, e_i = 1 Vrms @ f = 2400 Hz	I_L = -125mA	T_C = 25°C 4/ (figure 14)	45	-	dB
Output noise voltage	V_{NO}	V_{IN} = 10V	I_L = -50mA	T_C = 25°C 4/ (figure 15)	-	125	µVrms
Line transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	V_{IN} = 10V, ΔV_{IN} = 3.0V	I_L = -5mA	T_C = 25°C 5/ (figure 16)	-	30	mV/V
Load transient response	$\frac{\Delta V_{OUT}}{\Delta I_L}$	V_{IN} = 10V	I_L = -50mA, ΔI_L = -200mA	T_C = 25°C 5/ (figure 17)	-	2.5	mV/mA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 02 (see 3.4 unless otherwise specified) 1/2/-Continued.

Characteristic	Symbol	Condition: (fig. 12 unless otherwise stated)			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Output voltage	V_{OUT}	V_{IN} = 8V	I_L = -5mA, -1A		4.75	5.25	V
		V_{IN} = 30V	I_L = -5mA,-100mA		4.75	5.25	V
		V_{IN} = 38V	I_L = -1A		28.5	31.5	V
		V_{IN} = 10V	I_L = -5mA	T_C = 150°C	4.75	5.25	V
Line regulation	V_{RLINE}	8V < V_{IN} < 30V	I_L = -100mA		-150	150	mV
		8V < V_{IN} < 25V	I_L = -500mA		-50	50	mV
Load regulation	V_{RLOAD}	V_{IN} = 10V	-1A ≤ I_L ≤ -5mA		-100	100	mV
		V_{IN} = 30V	-100mA ≤ I_L ≤ -5mA		-150	150	mV
Thermal regulation	V_{RTH}	V_{IN} = 15V	I_L = -1A	T_C = 25°C	-50	50	mV
Standby current drain	I_{SCD}	V_{IN} = 10V	I_L = -5mA		-7.0	-0.5	mA
		V_{IN} = 30V	I_L = -5mA		-8.0	-0.5	mA
Standby current drain change versus line voltage	ΔI_{SCD} (LINE)	8V ≤ V_{IN} ≤ 30V	I_L = -5mA		-1.0	1.0	mA
Standby current drain change versus load current	ΔI_{SCD} (LOAD)	V_{IN} = 10V	-1A ≤ I_L ≤ -5mA		-0.5	0.5	mA
Control pin current	I_{CTL}	V_{IN} = 10V	I_L = -500mA	T_C = 25°C	-5.0	-0.01	µA
				-55°C ≤ T_C ≤ 125°C	-8.0	-0.01	µA
Output short circuit current	I_{OS1}	V_{IN} = 10V			-4.0	-1.00	A
	I_{OS2}	V_{IN} = 30V			-2.0	-0.02	A
Output voltage recovery after output short circuit current	V_{OUT} (RECOV)	V_{IN} = 10V	R_L = 5Ω, C_L = 20 µF	after I_{OS1} 3/	4.75	5.25	V
		V_{IN} = 30V	R_L = 5 kΩ	after I_{OS2} 3/	4.75	5.25	V
Voltage start-up	V_{START}	V_{IN} = 8V	R_L = 5Ω, C_L = 20µF		4.75	5.25	V
Ripple rejection	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	V_{IN} = 10V, e_i = 1 Vrms @ f = 2400 Hz	I_L = -350mA	T_C = 25°C 4/ (figure 14)	45	-	dB
Output noise voltage	V_{NO}	V_{IN} = 10V	I_L = -100mA	T_C = 25°C 4/ (figure 15)	-	250	µVrms
Line transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	V_{IN} = 10V, ΔV_{IN} = 3.0V	I_L = -5mA	T_C = 25°C 5/ (figure 16)	-	30	mV/V
Load transient response	$\frac{\Delta V_{OUT}}{\Delta I_L}$	V_{IN} = 10V	I_L = -100mA, ΔI_L = -400mA	T_C = 25°C 5/ (figure 17)	-	2.5	mV/mA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 03 (see 3.4 unless otherwise specified) 1/2/-Continued.

Characteristic	Symbol	Condition: (fig. 12 unless otherwise stated) Input voltage	Load current	Other	Limits Min	Limits Max	Unit
Output voltage	V_{OUT}	V_{IN} = 4.25V	I_L = -5mA, -500mA		1.20	1.30	V
		V_{IN} = 41.25V	I_L = -5mA, -50mA		1.20	1.30	V
		V_{IN} = 6.25V	I_L = -5mA	T_C = 150°C	1.20	1.30	V
Line regulation	V_{RLINE}	4.25V ≤ V_{IN} ≤ 41.25V	I_L = -5mA	T_C = 25°C	-9	9	mV
				-55°C ≤ T_C ≤ 125°C	-23	23	mV
Load regulation	V_{RLOAD}	V_{IN} = 6.25V	-500mA ≤ I_L ≤ -5mA	T_C = 25°C	-3.5	3.5	mV
				-55°C ≤ T_C ≤ 125°C	-12	12	mV
		V_{IN} = 41.25V	-50mA ≤ I_L ≤ -5mA	T_C = 25°C	-3.5	3.5	mV
				-55°C ≤ T_C ≤ 125°C	-12	12	mV
Thermal regulation	V_{RTH}	V_{IN} = 14.6V	I_L = -750mA	T_C = 25°C	-5	5	mV
Adjust pin current	I_{ADJ}	V_{IN} = 4.25V	I_L = -5mA		-100	-15	µA
		V_{IN} = 41.25V	I_L = -5mA		-100	-15	µA
Adjust pin current change versus line voltage	ΔI_{ADJ} (LINE)	4.25V ≤ V_{IN} ≤ 41.25V	I_L = -5mA		-5	5	µA
Adjust pin current change versus load current	ΔI_{ADJ} (LOAD)	V_{IN} = 6.25V	-500mA ≤ I_L ≤ -5mA		-5	5	µA
Minimum load current	I_Q	4.25V ≤ V_{IN} ≤ 14.25V forced V_{OUT} = 1.4V			-3.00	-0.50	mA
		V_{IN} = 41.25V forced V_{OUT} = 1.4V			-5.00	-1.00	mA
Output short circuit current	I_{OS1}	V_{IN} = 4.25V			-1.8	-0.50	A
	I_{OS2}	V_{IN} = 40V			-0.5	-0.05	A
Output voltage recovery after output short circuit current	V_{OUT} (RECOV)	V_{IN} = 4.25V	R_L = 2.5Ω, C_L = 20 µF	after I_{OS1} 3/	1.20	1.30	V
		V_{IN} = 40V	R_L = 250Ω	after I_{OS2} 3/	1.20	1.30	V
Voltage start-up	V_{START}	V_{IN} = 4.25V	R_L = 2.5Ω, C_L = 20µF		1.20	1.30	V
Ripple rejection	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	V_{IN} = 6.25V e_i = 1 Vrms @ f = 2400 Hz	I_L = -125mA	T_C = 25°C 4/ (figure 14)	65	-	dB
Output noise voltage	V_{NO}	V_{IN} = 6.25V	I_L = -50mA	T_C = 25°C 4/ (figure 15)	-	120	µVrms
Line transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	V_{IN} = 6.25V ΔV_{IN} = 3.0V	I_L = -10mA	T_C = 25°C 5/ (figure 16)	-	6	mV/V
Load transient response	$\frac{\Delta V_{OUT}}{\Delta I_L}$	V_{IN} = 6.25V	I_L = -50mA ΔI_L = -200mA	T_C = 25°C 5/ (figure 17)	-	0.6	mV/mA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 04 (see 3.4 unless otherwise specified) 1/2/-Continued.

Characteristic	Symbol	Condition: (fig. 12 unless otherwise stated)			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Output voltage	V_{OUT}	V_{IN} = 4.25V	I_L = -5mA, -1.5A		1.20	1.30	V
		V_{IN} = 41.25V	I_L = -5mA, -200mA		1.20	1.30	V
		V_{IN} = 6.25V	I_L = -5mA	T_C = 150°C	1.20	1.30	V
Line regulation	V_{RLINE}	4.25V ≤V_{IN} ≤41.25V	I_L = -5mA	T_C = 25°C	-9	9	mV
				-55°C ≤T_C ≤125°C	-23	23	mV
Load regulation	V_{RLOAD}	V_{IN} = 6.25V	-1.5A ≤I_L ≤-5mA	T_C = 25°C	-3.5	3.5	mV
				-55°C ≤T_C ≤125°C	-12	12	mV
		V_{IN} = 41.25V	-150mA ≤I_L ≤-5mA	T_C = 25°C	-3.5	3.5	mV
				-55°C ≤T_C ≤125°C	-12	12	mV
Thermal regulation	V_{RTH}	V_{IN} = 14.6V	I_L = -1.5A	T_C = 25°C	-5	5	mV
Adjust pin current	I_{ADJ}	V_{IN} = 4.25V	I_L = -5mA		-100	-15	µA
		V_{IN} = 41.25V	I_L = -5mA		-100	-15	µA
Adjust pin current change versus line voltage	ΔI_{ADJ} (LINE)	4.25V ≤V_{IN} ≤41.25V	I_L = -5mA		-5	5	µA
Adjust pin current change versus load current	ΔI_{ADJ} (LOAD)	V_{IN} = 6.25V	-1.5A ≤I_L ≤-5mA		-5	5	µA
Minimum load current	I_Q	4.25V ≤V_{IN} ≤14.25V forced V_{OUT} = 1.4V			-3.00	-0.50	mA
		V_{IN} = 41.25V forced V_{OUT} = 1.4V			-5.00	-1.00	mA
Output short circuit current	I_{OS1}	V_{IN} = 4.25V			-3.50	-1.50	A
	I_{OS2}	V_{IN} = 40V			-1.00	-0.18	A
Output voltage recovery after output short circuit current	V_{OUT} (RECOV)	V_{IN} = 4.25V	R_L = 0.833Ω, C_L = 20 µF	after I_{OS1} 3/	1.20	1.30	V
		V_{IN} = 40V	R_L = 250Ω	after I_{OS2} 3/	1.20	1.30	V
Voltage start-up	V_{START}	V_{IN} = 4.25V	R_L = 0.833Ω, C_L = 20µF		1.20	1.30	V
Ripple rejection	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	V_{IN} = 6.25V e_i = 1 Vrms @ f = 2400 Hz	I_L = -500mA	T_C = 25°C 4/ (figure 14)	65	-	dB
Output noise voltage	V_{NO}	V_{IN} = 6.25V	I_L = -100mA	T_C = 25°C 4/ (figure 15)	-	120	µVrms
Line transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	V_{IN} = 6.25V ΔV_{IN} = 3.0V	I_L = -10mA	T_C = 25°C 5/ (figure 16)	-	6	mV/V
Load transient response	$\frac{\Delta V_{OUT}}{\Delta I_L}$	V_{IN} = 6.25V	I_L = -100mA ΔI_L = -400mA	T_C = 25°C 5/ (figure 17)	-	0.30	mV/mA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 05 (see 3.4 unless otherwise specified) 1/2/-Continued.

Characteristic	Symbol	Condition: (fig. 13 unless otherwise stated)			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Output voltage	V_{OUT}	V_{IN} = 4.25V	I_L = -5mA, -3.0A		1.20	1.30	V
		V_{IN} = 36.25V	I_L = -5mA, -150mA		1.20	1.30	V
		V_{IN} = 6.25V	I_L = -5mA	T_C = 150°C	1.20	1.30	V
Line regulation	V_{RLINE}	4.25V ≤ V_{IN} ≤ 36.25V	I_L = -5mA	T_C = 25°C	-4	4	mV
				-55°C ≤ T_C ≤ 125°C	-20	20	mV
Load regulation	V_{RLOAD}	V_{IN} = 6.25V	-3.0A ≤ I_L ≤ -5mA	T_C = 25°C	-3.5	3.5	mV
				-55°C ≤ T_C ≤ 125°C	-12	12	mV
		V_{IN} = 36.25V	-150mA ≤ I_L ≤ -5mA	T_C = 25°C	-3.5	3.5	mV
				-55°C ≤ T_C ≤ 125°C	-12	12	mV
Thermal regulation	V_{RTH}	V_{IN} = 11.25V	I_L = -1.0A	T_C = 25°C	-5	5	mV
Adjust pin current	I_{ADJ}	V_{IN} = 4.25V	I_L = -5mA		-100	-15	µA
		V_{IN} = 36.25V	I_L = -5mA		-100	-15	µA
Adjust pin current change versus line voltage	ΔI_{ADJ} (LINE)	4.25V ≤ V_{IN} ≤ 36.25V	I_L = -5mA		-5	5	µA
Adjust pin current change versus load current	ΔI_{ADJ} (LOAD)	V_{IN} = 6.25V	-3.0A ≤ I_L ≤ -5mA		-5	5	µA
Minimum load current	I_Q	4.25V ≤ V_{IN} ≤ 14.25V forced V_{OUT} = 1.4V			-3.00	-0.50	mA
		V_{IN} = 36.25V forced V_{OUT} = 1.4V			-5.00	-0.50	mA
Output short circuit current	I_{OS1}	V_{IN} = 4.25V			-5.2	-3.0	A
	I_{OS2}	V_{IN} = 35V			-2.0	-0.15	A
Output voltage recovery after output short circuit current	V_{OUT} (RECOV)	V_{IN} = 4.25V	R_L = 0.416Ω, C_L = 20 µF	after I_{OS1} 3/	1.20	1.30	V
		V_{IN} = 35V	R_L = 250Ω	after I_{OS2} 3/	1.20	1.30	V
Voltage start-up	V_{START}	V_{IN} = 4.25V	R_L = 0.416Ω, C_L = 20µF		1.20	1.30	V
Ripple rejection	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	V_{IN} = 6.25V e_i = 1 Vrms @ f = 2400 Hz	I_L = -500mA	T_C = 25°C 4/ (figure 14)	65	-	dB
Output noise voltage	V_{NO}	V_{IN} = 6.25V	I_L = -100mA	T_C = 25°C 4/ (figure 15)	-	120	µVrms
Line transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	V_{IN} = 6.25V ΔV_{IN} = 3.0V	I_L = -10mA	T_C = 25°C 5/ (figure 16)	-	12	mV/V
Load transient response	$\frac{\Delta V_{OUT}}{\Delta I_L}$	V_{IN} = 6.25V	I_L = -100mA ΔI_L = -400mA	T_C = 25°C 5/ (figure 17)	-	0.30	mV/mA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 06 (see 3.4 unless otherwise specified) 1/2/-Continued.

Characteristic	Symbol	Condition: (fig. 13 unless otherwise stated)			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Output voltage	V_{OUT}	V_{IN} = 4.25V	I_L = -5mA, -5.0A		1.19	1.29	V
		V_{IN} = 36.25V	I_L = -5mA, -150mA		1.19	1.29	V
		V_{IN} = 6.25V	I_L = -7.0A		1.19	1.29	V
		V_{IN} = 6.25V	I_L = -5mA	T_C = 150°C	1.19	1.29	V
Line regulation	V_{RLINE}	4.25V ≤ V_{IN} ≤ 36.25V	I_L = -5mA	T_C = 25°C	-4	4	mV
				-55°C ≤ T_C ≤ 125°C	-17	17	mV
Load regulation	V_{RLOAD}	V_{IN} = 6.25V	-5.0A ≤ I_L ≤ -5mA	T_C = 25°C	-3.8	3.8	mV
				-55°C ≤ T_C ≤ 125°C	-8	8	mV
		V_{IN} = 36.25V	-150mA ≤ I_L ≤ -5mA	T_C = 25°C	-3.8	3.8	mV
				-55°C ≤ T_C ≤ 125°C	-8	8	mV
Thermal regulation	V_{RTH}	V_{IN} = 11.25V	I_L = -1.0A	T_C = 25°C	-2	2	mV
Adjust pin current	I_{ADJ}	V_{IN} = 4.25V	I_L = -5mA		-100	-15	µA
		V_{IN} = 36.25V	I_L = -5mA		-100	-15	µA
Adjust pin current change versus line voltage	ΔI_{ADJ} (LINE)	4.25V ≤ V_{IN} ≤ 36.25V	I_L = -5mA		-5	5	µA
Adjust pin current change versus load current	ΔI_{ADJ} (LOAD)	V_{IN} = 6.25V	-5.0A ≤ I_L ≤ -5mA		-5	5	µA
Minimum load current	I_Q	4.25V ≤ V_{IN} ≤ 14.25V forced V_{OUT} = 1.4V			-3.00	-0.50	mA
		V_{IN} = 36.25V forced V_{OUT} = 1.4V			-5.00	-0.50	mA
Output short circuit current	I_{OS1}	V_{IN} = 4.25V		t = 0.1 ms	-16.0	-7.0	A
	I_{OS2}	V_{IN} = 4.25V		t = 0.5 ms	-16.0	-7.0	A
	I_{OS3}	V_{IN} = 4.25V		t = 5.0 ms	-15.0	-5.0	A
	I_{OS4}	V_{IN} = 35 V		t = 10 ms	-3.0	-0.20	A
Output voltage recovery after output short circuit current	V_{OUT} (RECOV)	V_{IN} = 4.25V	R_L = 0.25Ω, C_L = 20 µF	after I_{OS3} 3/	1.19	1.29	V
		V_{IN} = 35V	R_L = 250Ω	after I_{OS4} 3/	1.19	1.29	V
Voltage start-up	V_{START}	V_{IN} = 4.25V	R_L = 0.25Ω, C_L = 20µF		1.19	1.29	V
Ripple rejection	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	V_{IN} = 6.25V e_i = 1 Vrms @ f = 2400 Hz	I_L = -500mA	T_C = 25°C 4/ (figure 14)	65	-	dB
Output noise voltage	V_{NO}	V_{IN} = 6.25V	I_L = -100mA	T_C = 25°C 4/ (figure 15)	-	120	µVrms
Line transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	V_{IN} = 6.25V ΔV_{IN} = 3.0V	I_L = -10mA	T_C = 25°C 5/ (figure 16)	-	12	mV/V
Load transient response	$\frac{\Delta V_{OUT}}{\Delta I_L}$	V_{IN} = 6.25V	I_L = -100mA ΔI_L = -400mA	T_C = 25°C 5/ (figure 17)	-	0.30	mV/mA

1/ All tests performed at T_C = 125°C may, at the manufacturer's option, be performed at T_C = 150°C. Specifications for T_C = 125°C shall then apply at T_C = 150°C.
2/ Static tests with load currents greater than 5 mA are performed under pulsed conditions defined on figure 12, or 13 as applicable.
3/ Output voltage recovery test shall be performed, with the designated load conditions, immediately after removal of each I_{OS} test forced output voltage condition.
4/ The meter for e_i and e_o shall have a minimum bandwidth from 10 Hz to 10 kHz and shall measure true rms voltages.
5/ The oscilloscope shall have a bandwidth between 5 and 15 MHz.

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3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Block diagrams and terminal connections. The block diagrams and terminal connections shall be as specified on figures 5 through 8.

3.2.2 Schematic circuits. The schematic circuits shall be as specified on figure 9.

3.2.3 Case outlines. The case outlines shall be as specified in 1.2.3 and on figures 1 through 4.

3.3 Lead material and lead finish.

3.3.1 Lead material. The lead material shall be in accordance with MIL-M-38510, or as follows:

Type X	
Nickel	50-53 percent
Manganese	0.60 percent, maximum
Silicon	0.30 percent, maximum
Carbon	0.10 percent, maximum
Chromium	0.25 percent, maximum
Cobalt	0.50 percent, maximum
Phosphorous	0.025 percent, maximum
Sulfur	0.025 percent, maximum
Aluminum	0.10 percent, maximum
Iron	Remainder

3.3.2 Lead finish. The lead finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The following electrical performance characteristics apply over the full operating case temperature range of -55°C to +125°C, unless otherwise specified in table I (see test conditions in 3.4.2).

3.4.1 Stability. If the device is located an appreciable distance from the power supply filter, a solid tantalum bypass capacitor should be connected as close to the device V_{CC} input as possible to suppress oscillation. A solid tantalum bypass capacitor is recommended on the device output. Since load currents of less than 5 milliamperes may result in a loss of voltage regulation, regulators should be preloaded with 5 milliamperes of load current in lightly loaded applications. In applications where fast rising high current pulses are present, additional output capacitance of 20 µF or more shall be used.

3.4.2 Test limits. The test limits specified in tables I and III apply only for the stated test conditions (e.g., 2 percent duty cycle), which essentially keep the junction temperature constant. In most applications the junction temperature will greatly exceed the 25°C ambient or sink temperature; thus devices may not perform within the 25°C specified limits.

3.5 Electrical test requirements. Electrical test requirements shall be as specified in table II. The subgroups of table III which constitute the minimum electrical test requirements for screening, qualification, and quality conformance by device class are specified in table II.

3.6 Marking. Marking shall be in accordance with MIL-M-38510 and 1.2. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 52 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

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TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)		
	Class S devices	Class B devices	Class C devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1	None
Final electrical test parameters (method 5004)	1*,2,3,4	1*,2,3,4	1,4
Group A test requirements (method 5005)	1,2,3,4,7	1,2,3,4,7	1,2,3,4,7
Group C end-point and group B, class S, electrical parameters (method 5005)	1,2,3, and table IV delta limits	1 and table IV delta limits	1 and table IV delta limits
Group D end-point electrical parameters (method 5005)	1,2,3	1	1

* PDA applies to subgroup 1 (see 4.2e).

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. For class S and B devices, an additional burn-in screen shall be performed in accordance with figure 10 to test the operation of the thermal shutdown circuit. This screen shall be performed after serialization (3.1.8 of method 5004 of MIL-STD-883) and before interim electrical parameters (pre burn-in, 3.1.9 of method 5004 of MIL-STD-883). The requirements of 3.2.3 of method 1015 of MIL-STD-883 shall apply to this screen except the devices need not be tested in an oven.

b. Interim electrical parameters shall be performed for class S and B devices and shall be as specified in table II.

c. Burn-in (method 1015 of MIL-STD-883). For class S and B devices, only test condition C shall apply using the circuit shown on figure 11. Accelerated test conditions shall not be used.

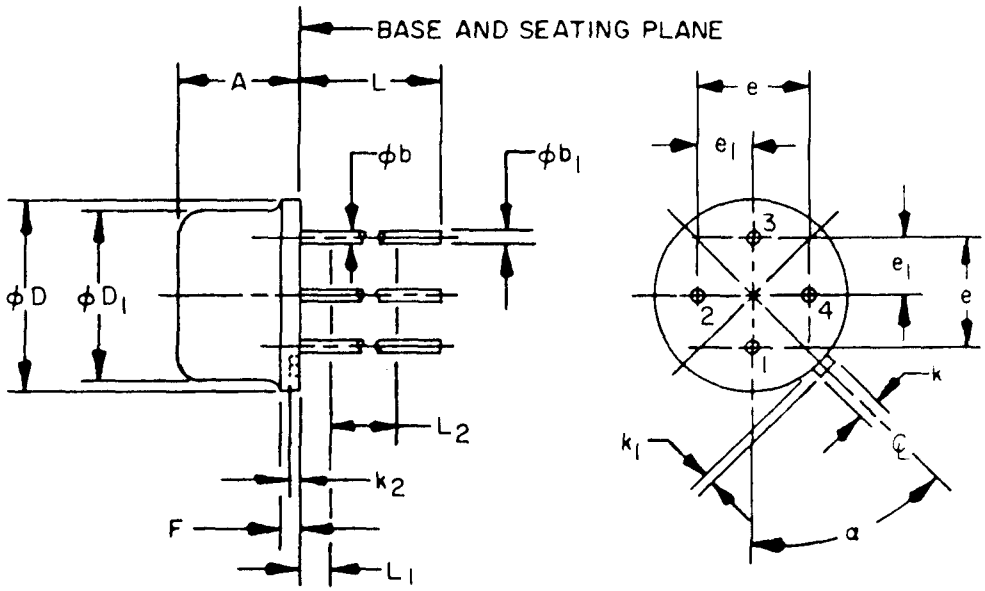
d. Reverse bias burn-in shall not be performed.

e. Percent defective allowable (PDA) - The PDA for class S devices shall be as specified in MIL-M-38510. The PDA is specified as 10 percent for class B devices based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883 and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

f. Constant acceleration (method 2001 of MIL-STD-883); test condition B shall be used for case Y.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

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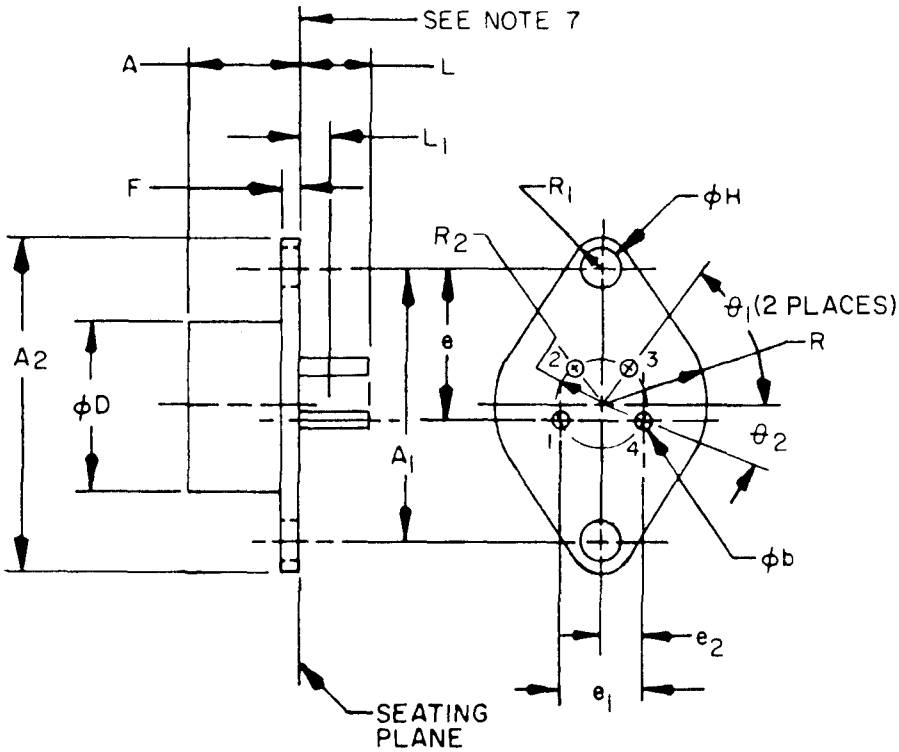


Symbol	Inches		Millimeters		Notes
	Min	Max	Min	Max	
A	.240	.260	6.10	6.60	
Øb	.016	.019	.41	.48	3
$Øb_1$	.016	.021	.41	.53	3
ØD	.335	.370	8.51	9.40	
$ØD_1$	.305	.335	7.75	8.51	
e	.200 T.P.		5.08 T.P.		5
e_1	.100 T.P.		2.54 T.P.		5
F	---	.050	---	1.27	
k	.028	.034	.71	.86	
k_1	.029	.045	.74	1.14	4
k_2	.009	.041	.23	1.04	
L	.500	---	12.70	---	
L_1	---	.050	---	1.27	
L_2	.250	---	6.35	---	
α	45° T.P.		45° T.P.		5

NOTES:
1. Dimensions shall be measured in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. (Four leads) Øb applies between L_1 and L_2. $Øb_1$ applies between L_2 and .500 (12.70 mm) from the reference plane. Diameter is uncontrolled in L_1 and beyond .500 (12.70 mm) from the reference plane.
4. Four leads.
5. Measured from the maximum diameter of the product.
6. Leads having a maximum diameter .019 (.48 mm) measured in gaging plane .054 (1.37 mm) + .001 (.03 mm) -.000 (.00 mm) below the base plane of the product shall be within .007 (.18 mm) of their true position relative to a maximum width tab.
7. The product may be measured by direct methods or by gage.

FIGURE 1. Case outline X (device type 01).

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.250	.450	6.35	11.43	
A_1	1.177	1.197	29.90	30.40	
A_2	1.480	1.500	37.59	38.10	
Øb	.038	.043	.97	1.09	3,7
ØD	---	.875	---	22.22	
e	.655	.675	16.64	17.14	
e_1	.420	.440	10.67	11.18	
e_2	.205	.225	5.21	5.72	
F	.060	.135	1.52	3.43	
ØH	.151	.161	3.84	4.09	5,6
L	.312	.500	7.92	12.70	4
L_1	---	.050	---	1.27	3,4
R	.495	.525	12.57	13.34	
R_1	.131	.188	3.33	1.78	
R_2	.470 T.P.		11.94 T.P.		
θ_1	54° T.P.		54° T.P.		
θ_2	18° T.P.		18° T.P.		

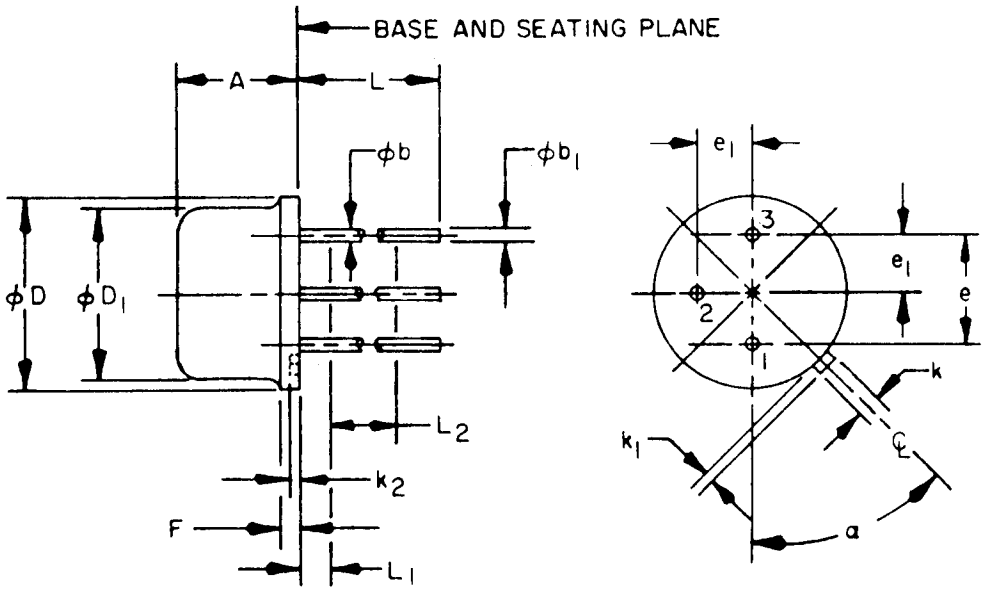
FIGURE 2. Case outline Y (device type 02).

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NOTES:
1. Dimensions shall be measured in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. (Four leads) Øb applies between L_1 and .500 (12.70 mm) from the seating plane. Diameter is uncontrolled in L_1 and beyond .500 (12.70 mm) from the seating plane.
4. Four leads.
5. Two holes.
6. Two holes located at true position within diameter .010 (.25 mm).
7. Leads having a maximum diameter of .043 (1.09 mm) measured in gaging plane .054 (1.37 mm) + .001 (.03 mm) -.000 (.00 mm) below the seating plane shall be located at true position within diameter .014 (.36 mm).
8. The mounting surface of the header shall be flat to convex within .003 (.08 mm) inside a .930 (23.62 mm) diameter circle on the center of the header and flat to convex within .006 (.15 mm) overall.

FIGURE 2. Case outline Y (device type 02) - Continued.

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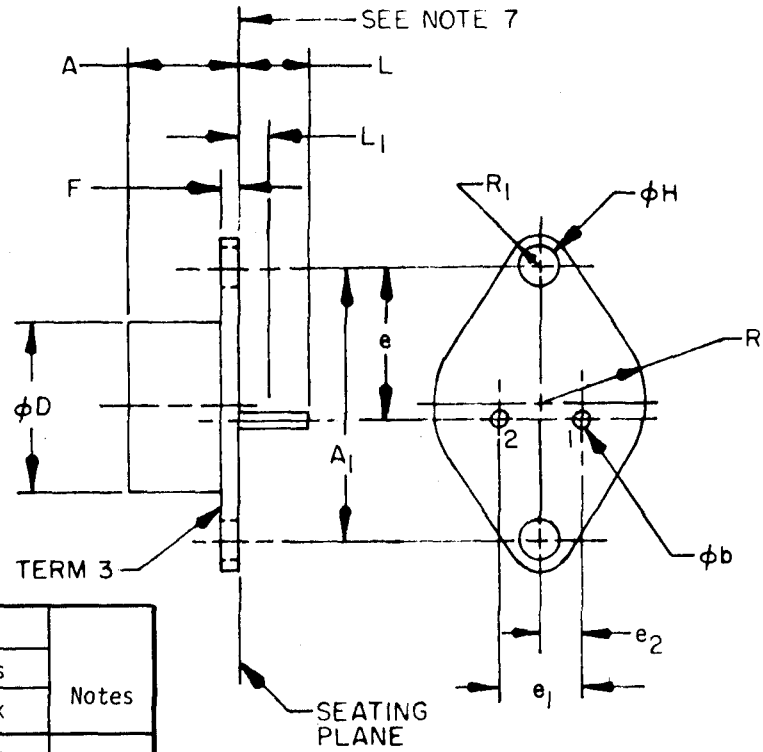


Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.165	.185	4.19	4.70	
Øb	.016	.019	.41	.48	3
$Øb_1$	.016	.021	.41	.53	3
ØD	.335	.370	8.51	9.40	
$ØD_1$	.305	.335	7.75	8.51	
e	.200 T.P.		5.08 T.P.		5
e_1	.100 T.P.		2.54 T.P.		5
F	---	.050	---	1.27	
k	.028	.034	.71	.86	
k_1	.029	.045	.74	1.14	4
k_2	.009	.041	.23	1.04	
L	.500	---	12.70	---	
L_1	---	.050	---	1.27	
L_2	.250	---	6.35	---	
α	45° T.P.		45° T.P.		5

NOTES:
1. Dimensions shall be measured in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. (Three leads) Øb applies between L_1 and L_2. $Øb_1$ applies between L_2 and .500 (12.70 mm) from the reference plane. Diameter is uncontrolled in L_1 and beyond .500 (12.70 mm) from the reference plane.
4. Three leads.
5. Measured from the maximum diameter of the product.
6. Leads having a maximum diameter .019 (.48 mm) measured in gaging plane .054 (1.37 mm) +.001 (.03 mm) -.000 (.00 mm) below the base plane of the product shall be within .007 (.18 mm) of their true position relative to a maximum width tab.
7. The product may be measured by direct methods or by gage.

FIGURE 3. Case outline X (device type 03).

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.250	.450	6.35	11.43	
A_1	1.177	1.197	29.90	30.40	
Øb	.038	.043	.97	1.09	3,7
ØD	---	.875	---	22.22	
e	.655	.675	16.64	17.14	
e_1	.420	.440	10.67	11.18	
e_2	.205	.225	5.21	5.72	
F	.060	.135	1.52	3.43	
ØH	.151	.161	3.84	4.09	5,6
L	.312	.500	7.92	12.70	4
L_1	---	.050	---	1.27	3,4
R	.495	.525	12.57	13.34	
R_1	.131	.188	3.33	4.78	

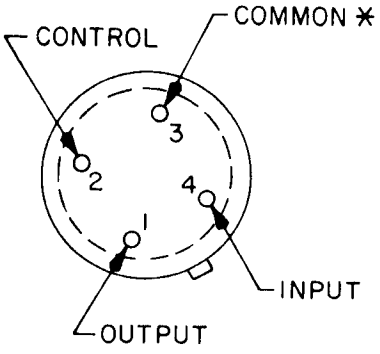
NOTES:
1. Dimensions shall be measured in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. (Two leads) Øb applies between L_1 and .500 (12.70 mm) from the seating plane. Diameter is uncontrolled in L_1 and beyond .500 (12.70 mm) from the seating plane.
4. Two leads.
5. Two holes.
6. Two holes located at true position within diameter .010 (.25 mm).
7. Leads having a maximum diameter of .043 (1.09 mm) measured in gaging plane .054 (1.37 mm) + .001 (.03 mm) - .000 (.00 mm) below the seating plane shall be located at true position within diameter .014 (.36 mm).
8. The mounting surface of the header shall be flat to convex within .003 (.08 mm) inside a .930 (23.62 mm) diameter circle on the center of the header and flat to convex within .006 (.15 mm) overall.

FIGURE 4. Case outline Y (device type 04, 05, and 06).

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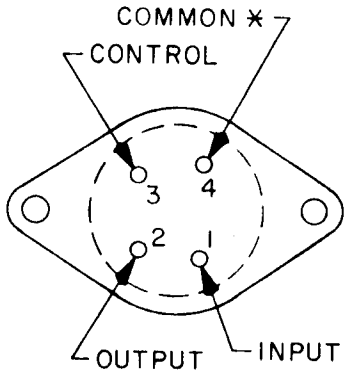
Case X

Device type 01



Case Y

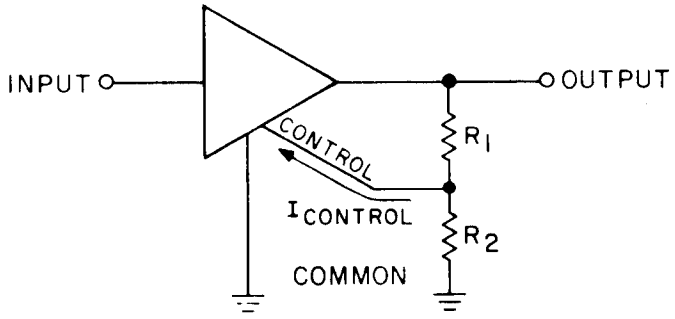
Device type 02



NOTE: Bottom views.

*(Case is connected to common).

FIGURE 5. Terminal connections for device types 01 and 02.



NOTES:

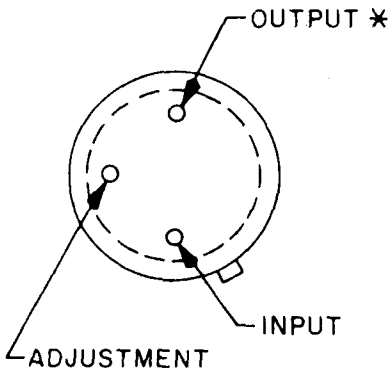
1. $V_{OUT} = \left[\left(\frac{R_1 + R_2}{R_2}\right)\left(V_{CONTROL}\right) + \left|I_{CONTROL}\right| \times R_1\right]$ Volts.
2. $V_{CONTROL}$ = 5.00 V (Nominal).
3. R_2 = 1.0 kΩ provides a minimum of |5 mA| load to the regulator at any V_{OUT}.

FIGURE 6. Block diagram for device types 01 and 02.

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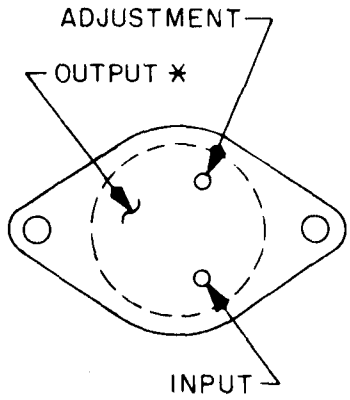
Case X

Device type 03



Case Y

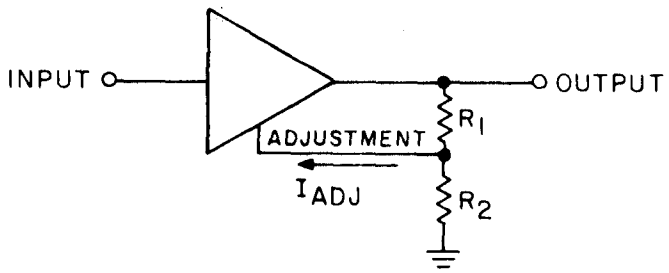
Device types 04, 05 and 06.



NOTE: Bottom views.

*(Case is connected to output).

FIGURE 7. Terminal connections for device types 03, 04, 05, and 06.



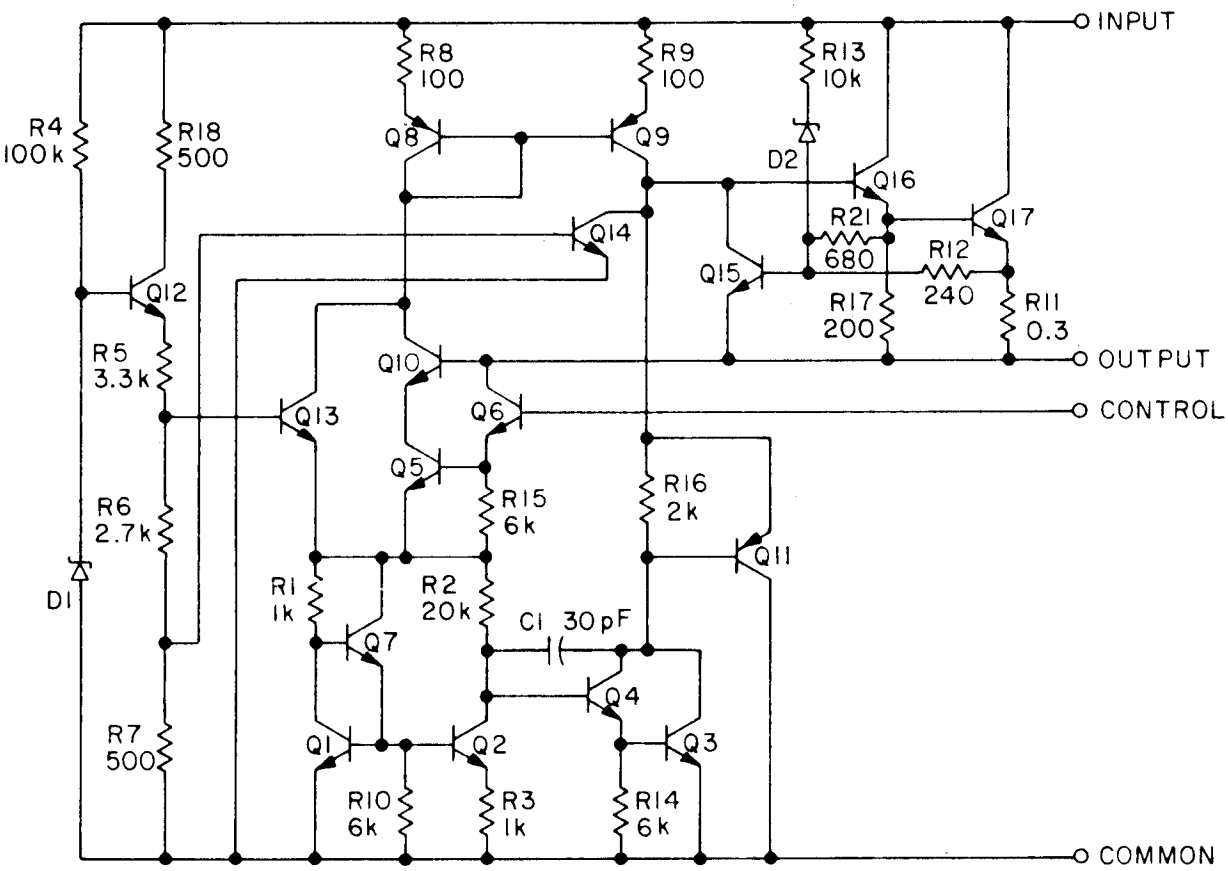
NOTES:

1. $V_{OUT} = \left[\left(\frac{R_1 + R_2}{R_1}\right)(1.25) + \left|I_{ADJ}\right| \times R_2\right]$ Volts.
2. R_1 = 250Ω provides a minimum of [5 mA] load to the regulator at any V_{OUT}.

FIGURE 8. Block diagram for device types 03, 04, 05, and 06.

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Device types 01 and 02



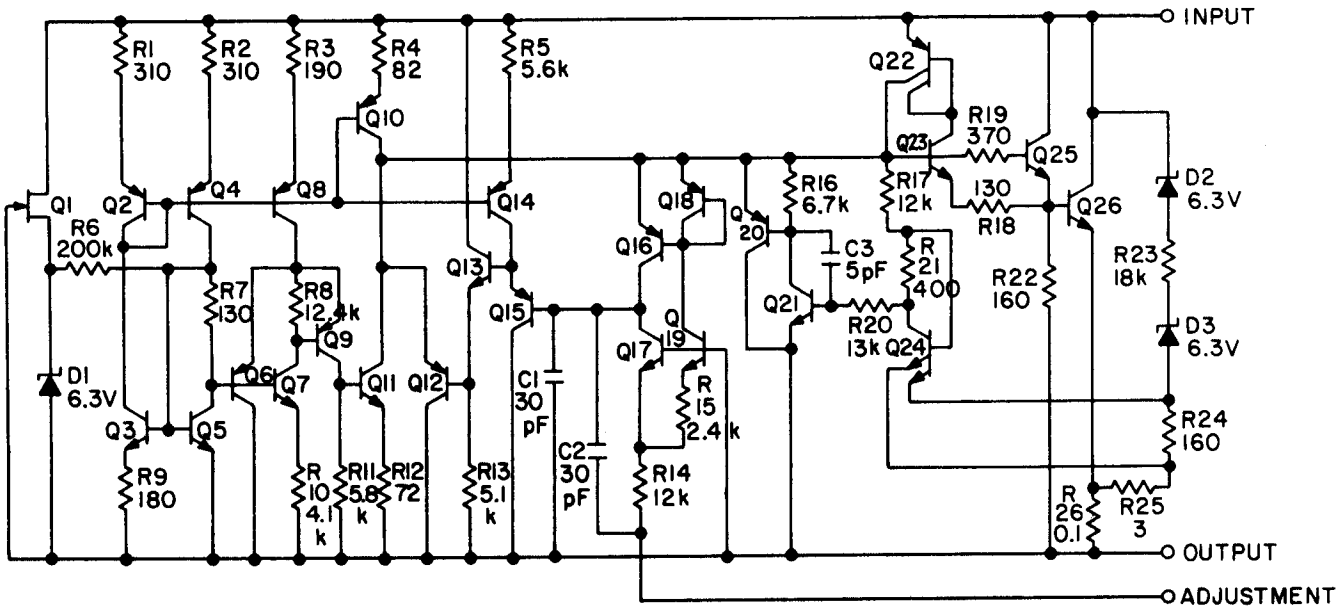
Fairchild Semiconductor

FIGURE 9. Schematic circuits.

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Device types 03 and 04

Circuit A



National Semiconductor

FIGURE 9. Schematic circuits - Continued.

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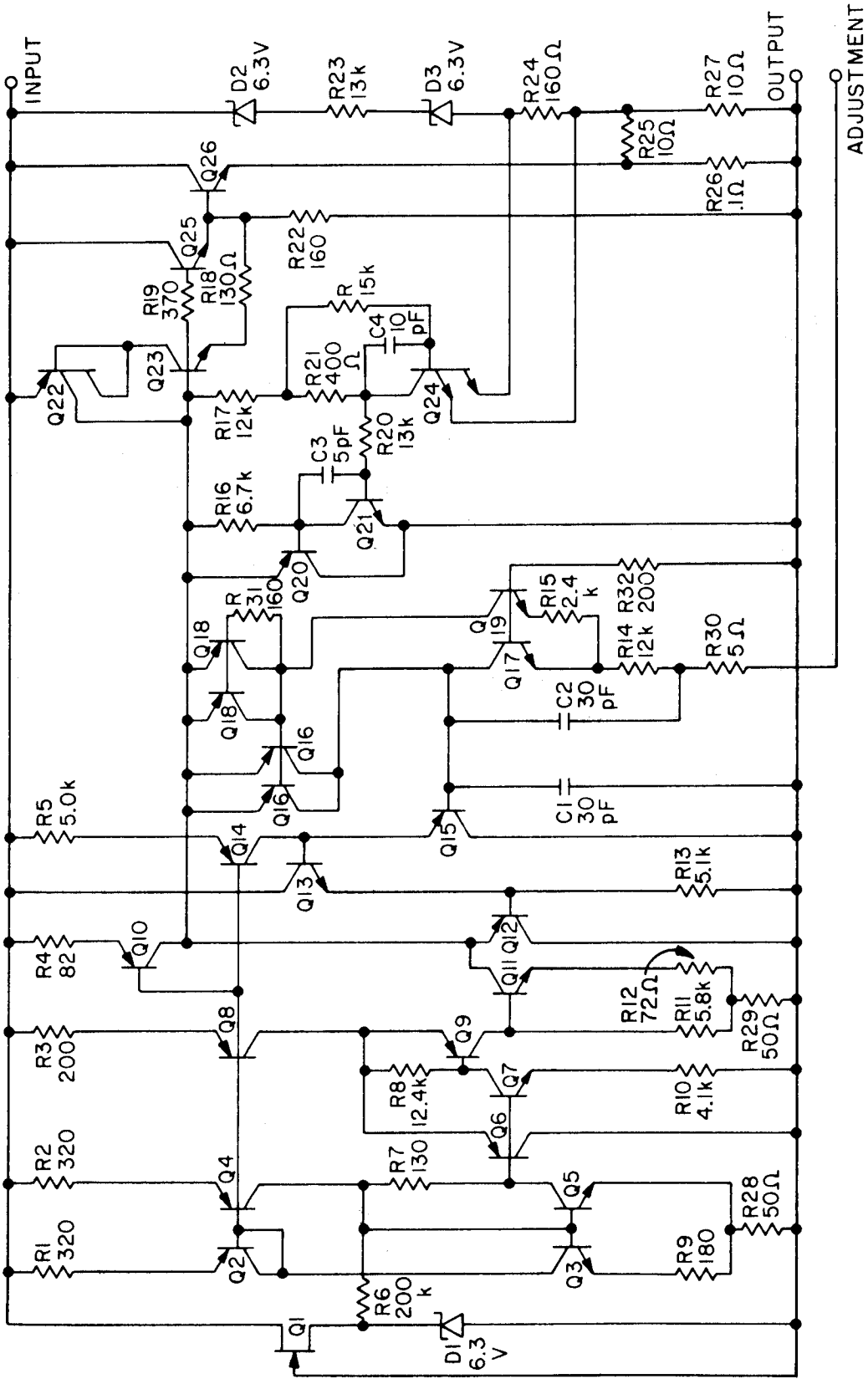
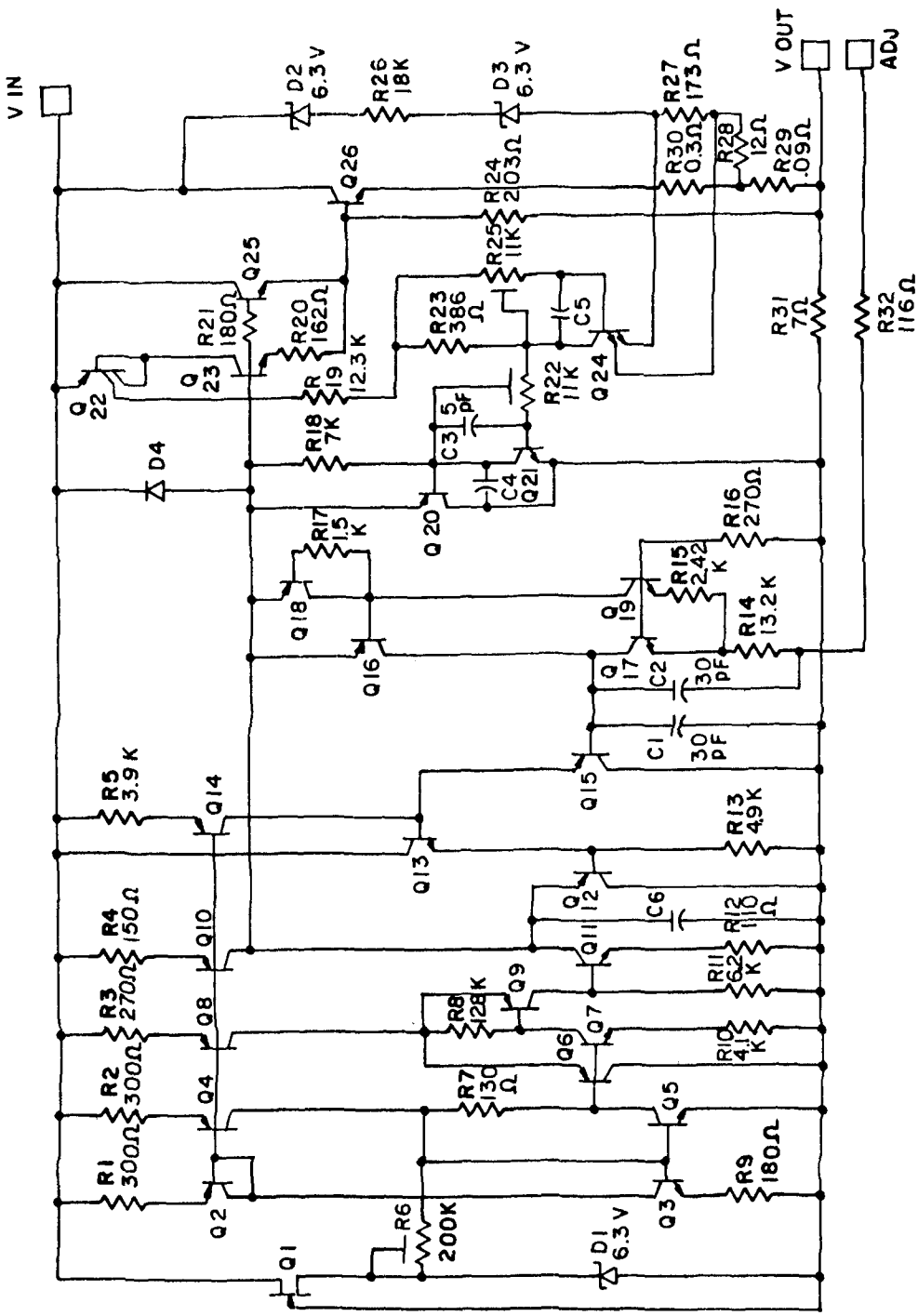


FIGURE 9. Schematic circuits - Continued.

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Device type 03 and 04

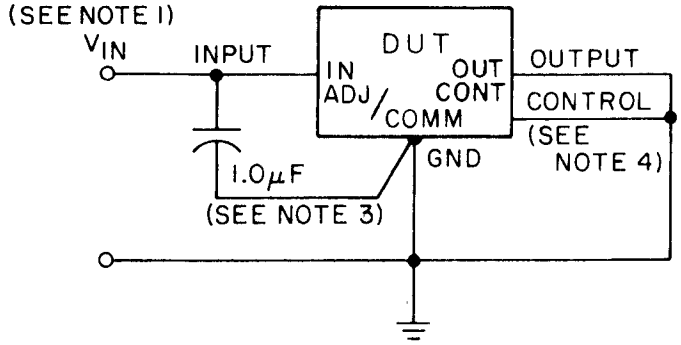
Circuit C



FAIRCHILD

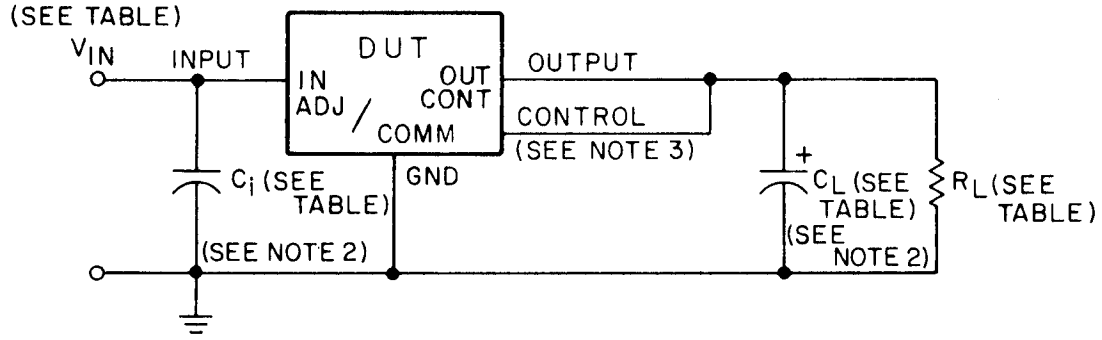
FIGURE 9. Schematic circuits - continued.

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NOTES:
1. For device types 01, 02, 03 and 04 - - - - V_{IN} = 40 V.
   For device types 05 and 06 - - - - - - - - V_{IN} = 35 V.
2. Test is conducted at T_A = 25°C without a heat sink for a minimum of 4 hours.
3. The 1.0 μF capacitor may be reduced or eliminated provided the device remains stable.
4. The control pin connection is required for device types 01 and 02 only.

FIGURE 10. Pre burn-in test circuit.



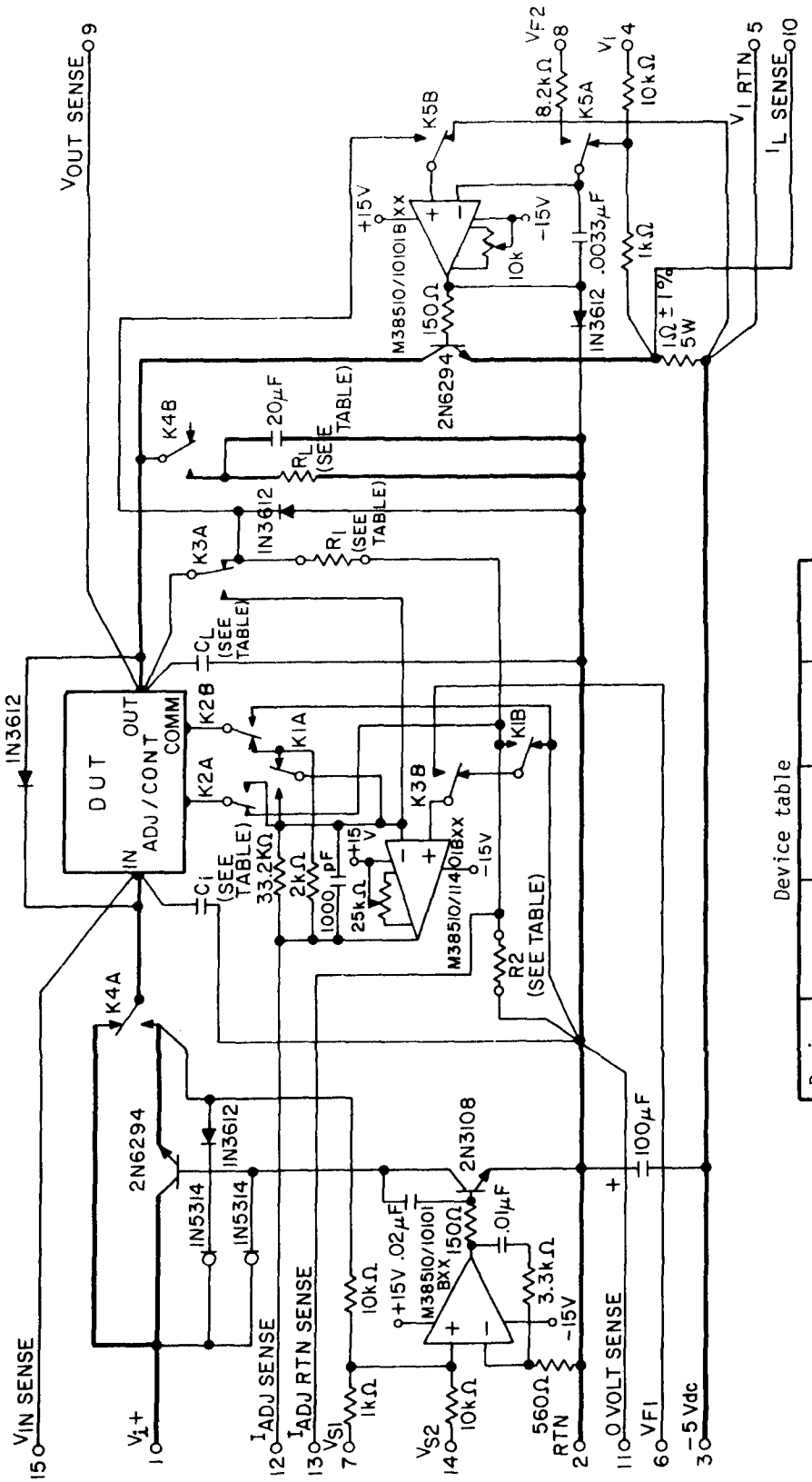
Device table

Device type	01	02	03	04	05	06
V_{IN}	20 V	20 V	36.5 V	36.5 V	36.5 V	36.5 V
R_L	887Ω	118Ω	250Ω	63.4Ω	63.4Ω	63.4Ω
C_I	0.33 μF	0.33 μF	1.0 μF	1.0 μF	1.0 μF	1.0 μF
C_L	0.1 μF	0.1 μF	1.0 μF	1.0 μF	1.0 μF	1.0 μF

NOTES:
1. Test is conducted without a heat sink.
2. The capacitors may be reduced or eliminated provided the device remains stable.
3. The control pin connection is required for device types 01 and 02 only.

FIGURE 11. Burn-in and operating life test circuit.

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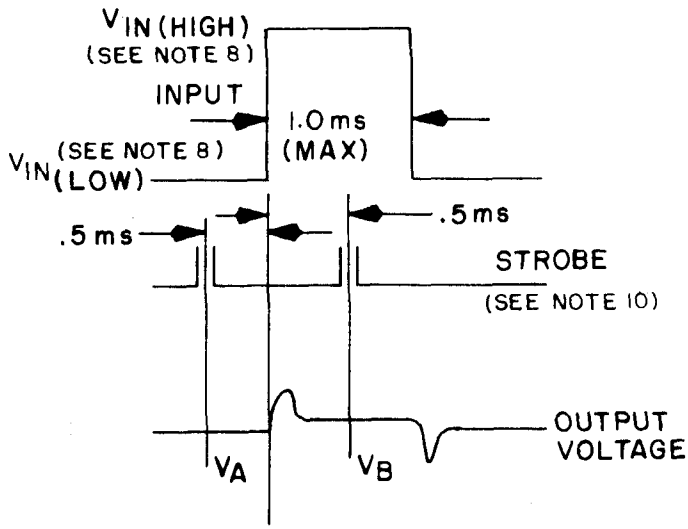
Device table

Device type	01	02	03	04
R_1	0	0	249Ω	249Ω
R_2	1KΩ	1KΩ	0	0
R_L	10Ω	5Ω	2.5Ω	0.833Ω
C_i	0.33 μF	0.33 μF	1.0 μF	1.0 μF
C_L	0.1 μF	0.1 μF	1.0 μF	1.0 μF

FIGURE 12. Test circuit for static tests for device types 01, 02, 03 and 04.

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LINE REGULATION WAVEFORMS



LOAD REGULATION WAVEFORMS

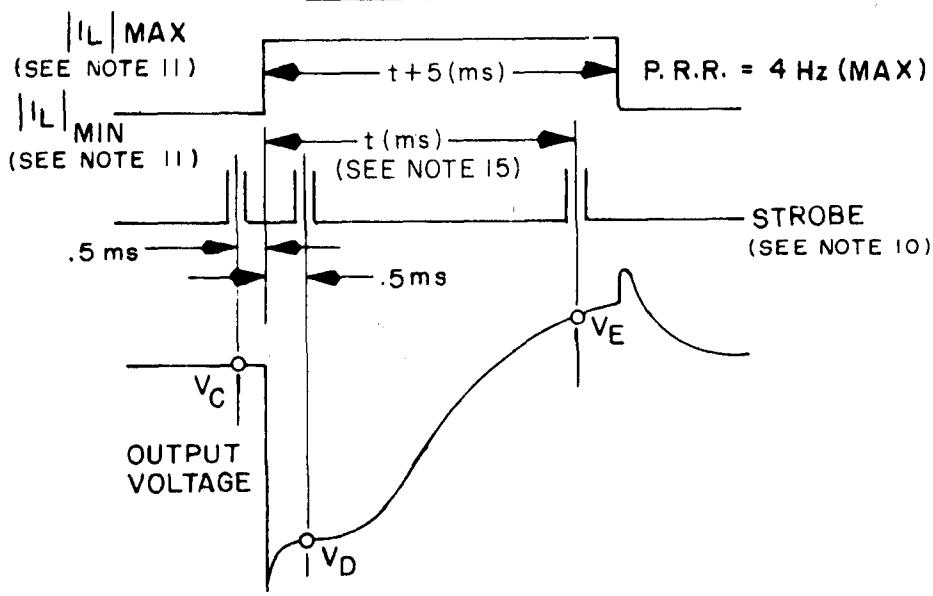


FIGURE 12. Test circuit static tests for device types 01 through 04 - Continued.

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Waveforms
for
voltage start-up,
output short circuit current,
and
output voltage recovery.

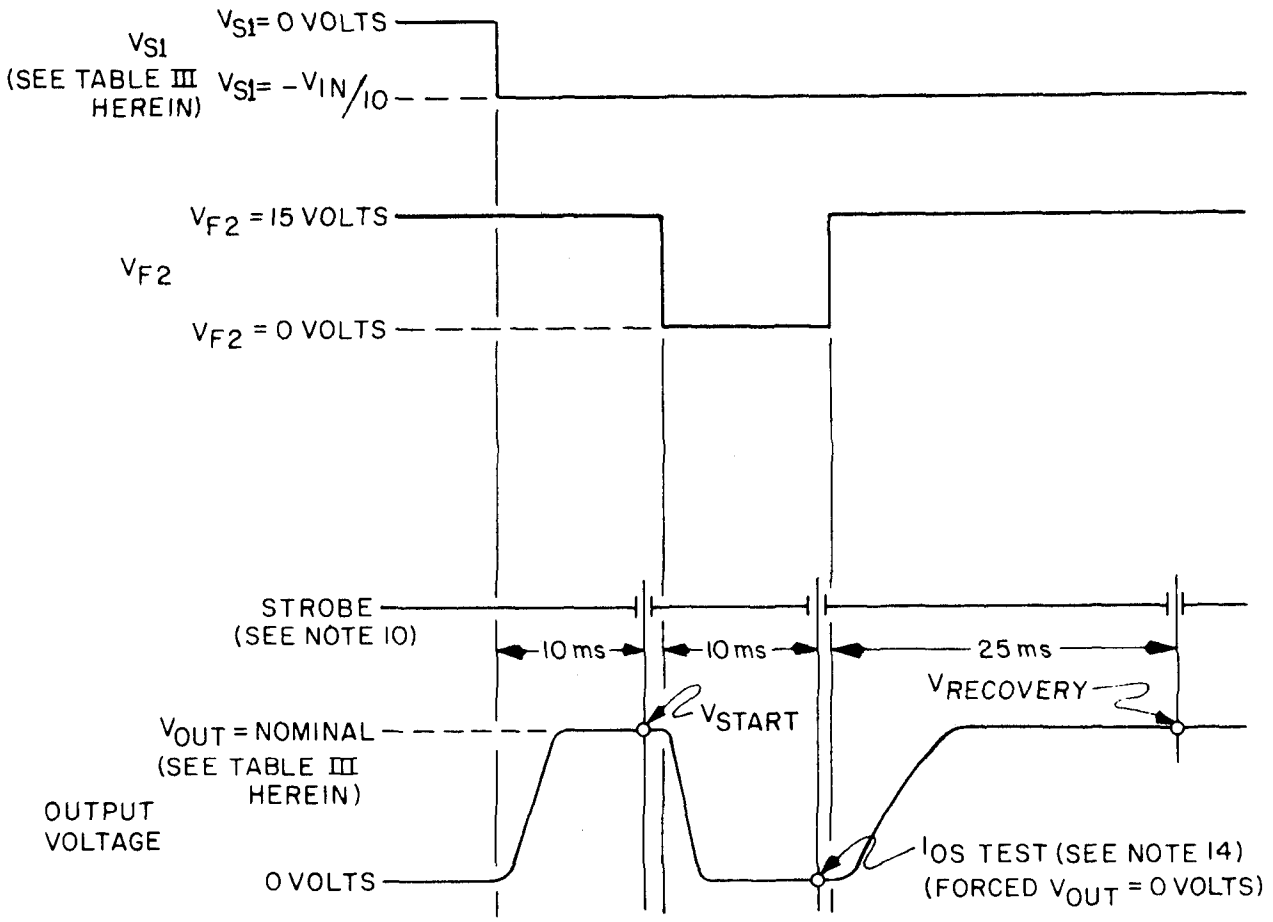


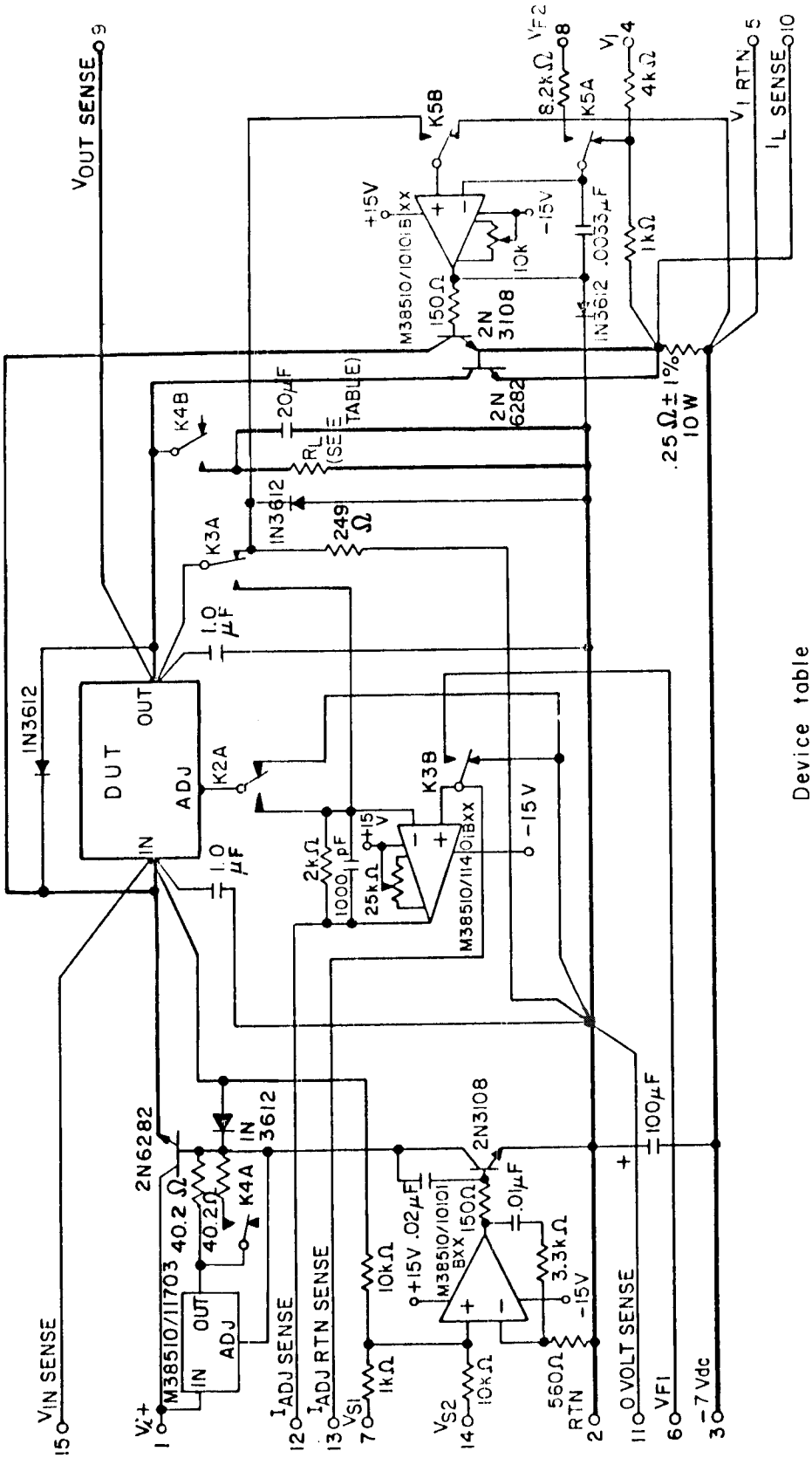
FIGURE 12. Test circuit for static tests for device types 01 through 04 - Continued.

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NOTES:
1. Heavy current paths ($I \geq 0.05A$) are indicated by bold lines.
2. Kelvin connections must be used for all output current and voltage measurements. For device types 03 and 04, output voltage measurements shall be made at the case.
3. The output offset voltage shall be adjusted to zero with the DUT removed. The operational amplifier stabilization networks may vary with test adapter construction. Alternate drive circuits for the 2N6294 may be used to develop the proper load current and input voltage pulses. These circuits shall require the approval of the qualifying activity.
4. Relay switch positions are defined in table III.
5. Load currents of 5 mA may be established via the load resistors R_1 and R_2. All other load currents shall be established via the pulse load circuit. Resistors R_1 and R_2 shall have a tolerance $\leq 0.1\%$ for device types 01 and 02.
6. The pulse generator for the pulse load circuit shall have the following characteristics.
   a. Pulse amplitude = $-10\ (|I_L|\ -V_0/(R_1 + R_2))$ volts. (referenced to -5 volts)
   b. Pulse width = 1.0 ms (unless otherwise stated).
   c. Duty cycle = 2% (maximum).
7. Load currents shall be determined by the voltage measured across the 1Ω resistor. Measurements shall be made 0.5 ms after the start of the pulse.
8. V_{IN} (LOW) and V_{IN} (HIGH) per table III herein.
9. $V_{RLINE} = V_B - V_A$.
10. The output voltage is sampled at specified intervals. Strobe pulse width is 100 µs maximum.
11. $|I_L|$ (minimum) and $|I_L|$ (maximum) per table III herein.
12. $V_{RLOAD} = V_D - V_C$.
13. $V_{RTH} = V_D - V_E$.
14. $I_{OS} = (I_L)$ amps.
15. For device types 01 and 02, t = 10.5 ms.
    For device types 03 and 04, t = 20.5 ms.

FIGURE 12. Test circuit for static test, for devices 01 thru 04 - cont.

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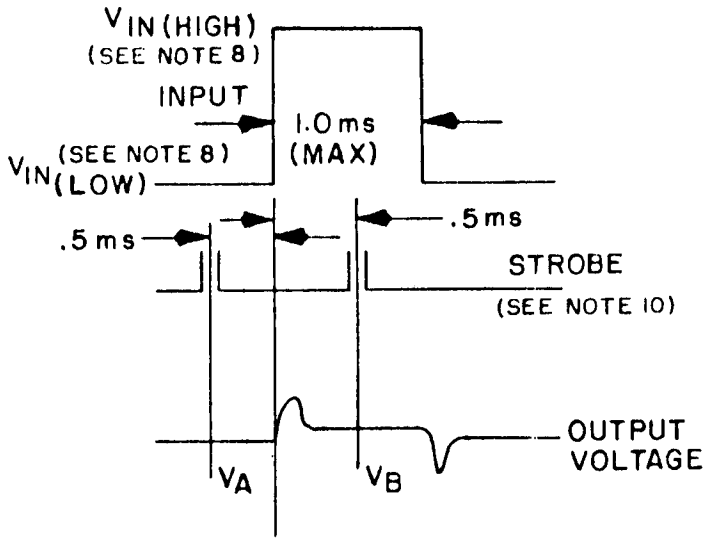
Device table

Device type	05	06
R_L	0.416Ω	0.25Ω

FIGURE 13. Test circuit for static tests for device types 05 and 06.

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LINE REGULATION WAVEFORMS



LOAD REGULATION WAVEFORMS

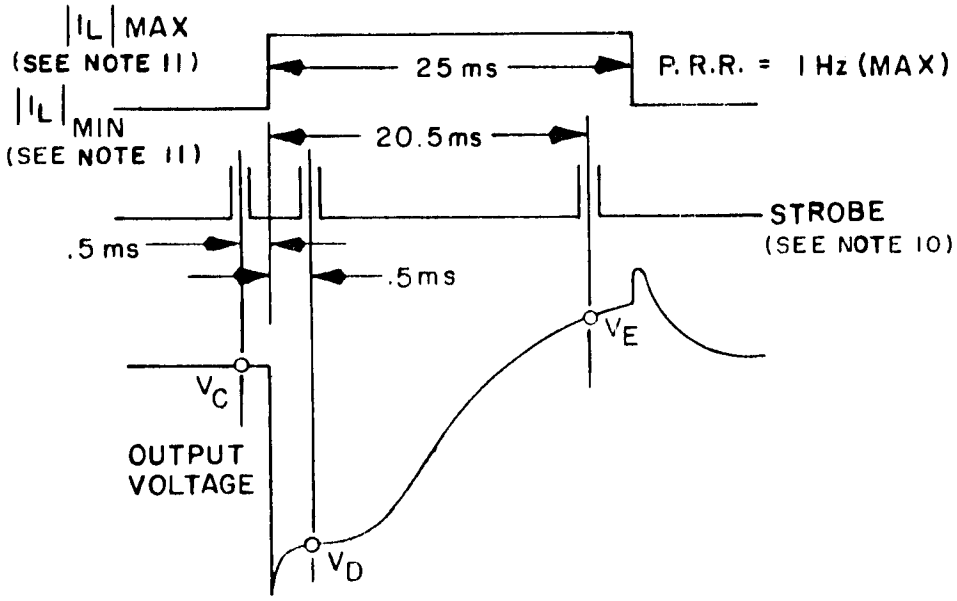


FIGURE 13. Test circuit for static tests for device types 05 and 06 - Continued.

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Waveforms
for
voltage start-up,
output short circuit current,
and
output voltage recovery.

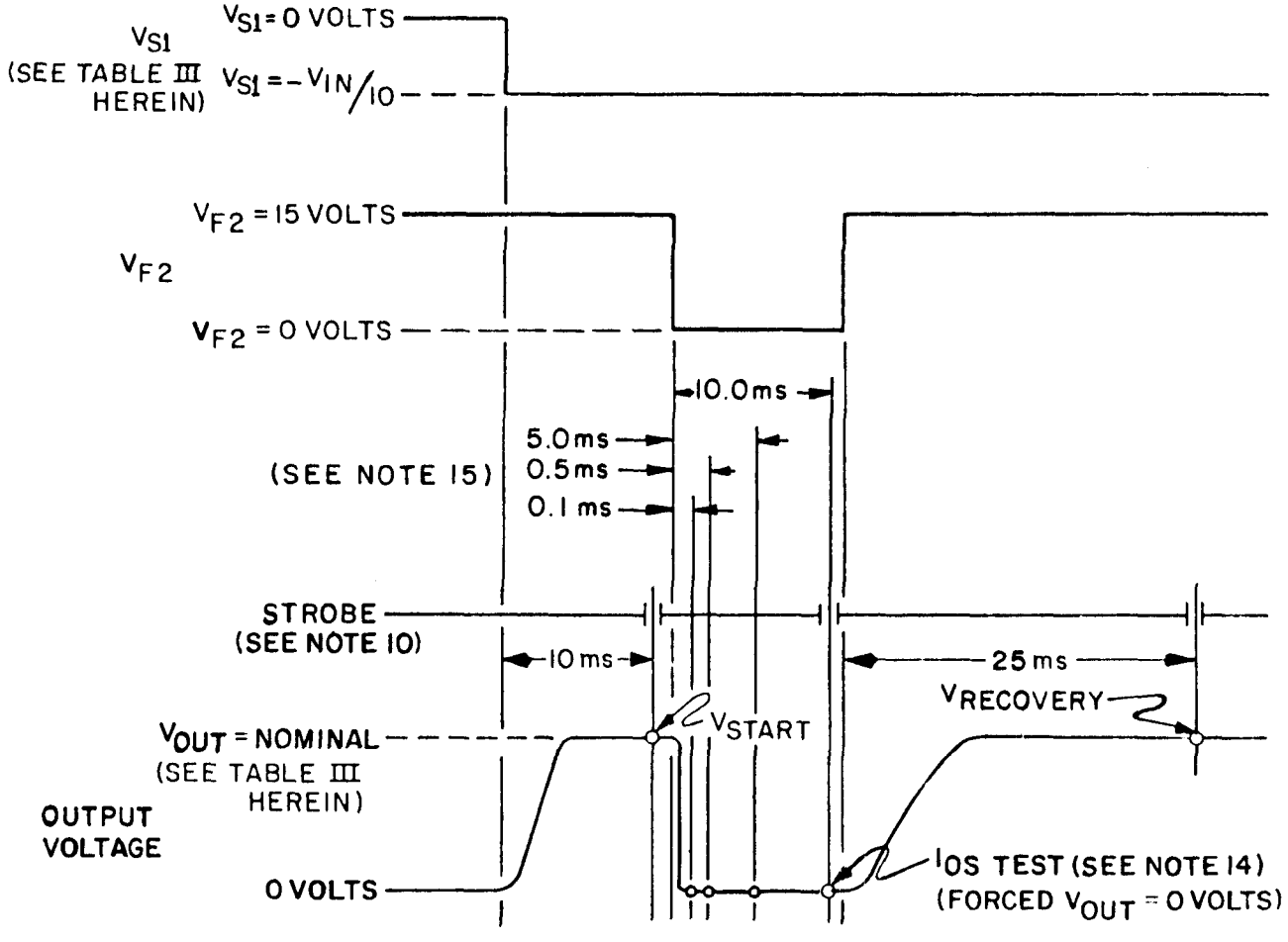


FIGURE 13. Test circuit for static tests for device types 05 and 06 - Continued.

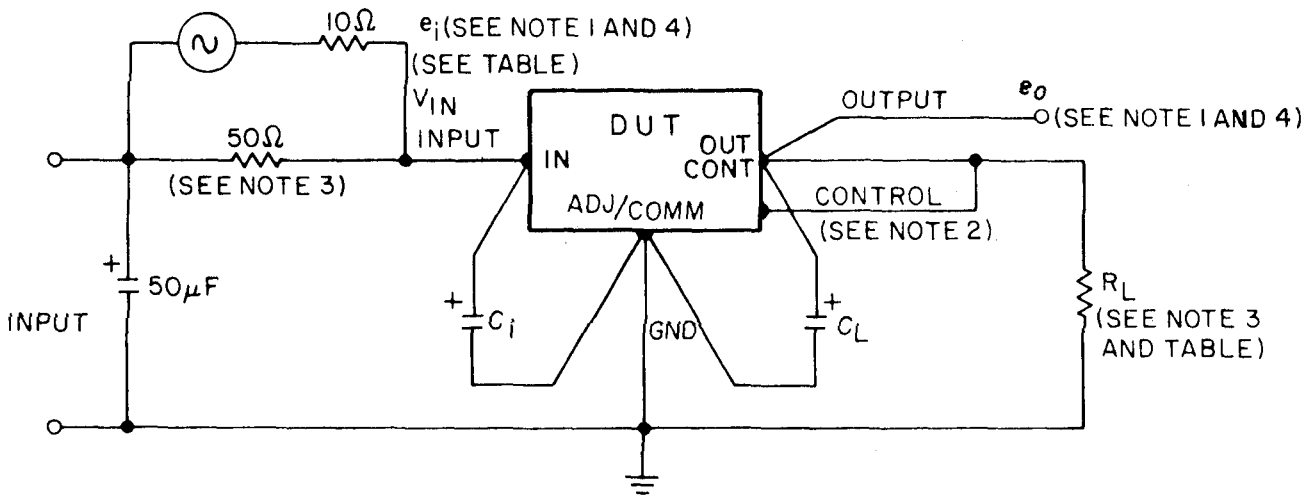
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NOTES:

1. Heavy current paths ($I \geq 0.1$ A) are indicated by bold lines.
2. Kelvin connections must be used for all output current and voltage measurements. For device types 05 and 06, output voltage measurements shall be made at the case.
3. The output offset voltage shall be adjusted to zero with the DUT removed. The operational amplifier stabilization networks may vary with test adapter construction. Alternate drive circuits for the 2N6282 may be used to develop the proper load current and input voltage pulses. These circuits shall require the approval of the qualifying activity.
4. Relay switch positions are defined in table III.
5. Load currents of 5 mA shall be established via the 249 Ω load resistor. All other load currents shall be established via the pulse load circuit.
6. The pulse generator for the pulse load circuit shall have the following characteristics.
   a. Pulse amplitude = $-(|I_L| - .005)$ volts. (referenced to -7 volts)
   b. Pulse width = 1.0 ms (unless otherwise stated).
   c. Duty cycle = 2% (maximum).
   d. Rise time = 30μ sec (minimum).
7. Load currents shall be determined by the voltage measured across the 0.25 Ω resistor. Measurements shall be made 0.5 ms after the start of the pulse.
8. V_{IN} (LOW) and V_{IN} (HIGH) per table III herein.
9. $V_{RLINE} = V_B - V_A$.
10. The output voltage is sampled at specified intervals. Strobe pulse width is 100 μs maximum.
11. $|I_L|$ (minimum) and $|I_L|$ (maximum) per table III herein.
12. $V_{RLOAD} = V_D - V_C$.
13. $V_{RTH} = V_D - V_E$.
14. $I_{OS} = (I_L)$ amps.
15. Output short circuit current measurements at t = 0.1, t = 0.5, and t = 5.0, are to be made on device type 06 only.

FIGURE 13. Test circuit for static tests for device types 05 and 06 - Continued.

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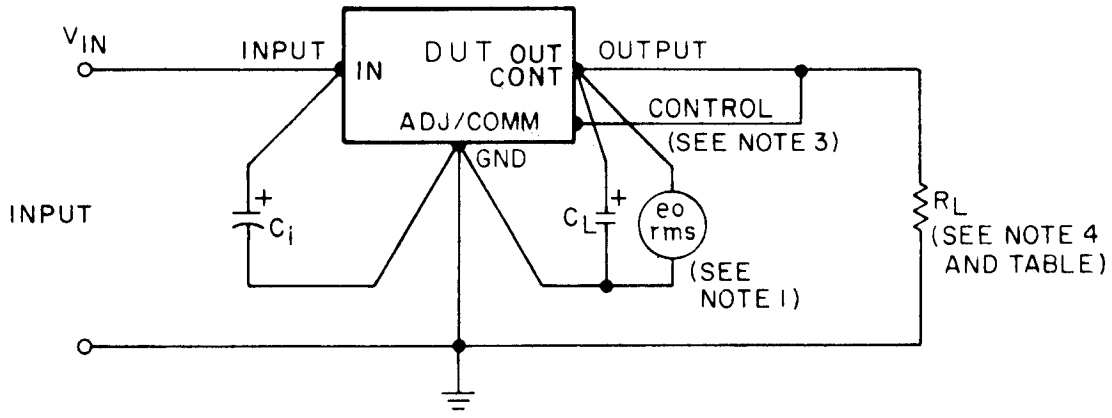
Device table

Device type	01	02	03	04	05	06
V_{IN}	10 V	10 V	6.25 V	6.25 V	6.25 V	6.25 V
R_L	40.2Ω	14.3Ω	10Ω	2.5Ω	2.5Ω	2.5Ω
C_i	0.33 μF	0.33 μF	1.0 μF	1.0 μF	1.0 μF	1.0 μF
C_L	0.1 μF	0.1 μF	1.0 μF	1.0 μF	1.0 μF	1.0 μF

NOTES:
1. e_i = 1 Vrms @ f = 2400 Hz (measured at the input terminals of the DUT).
   Ripple rejection = 20 log $\frac{e_{i\ rms}}{e_{o\ rms}}$.
2. The control pin connection is required for device types 01 and 02 only.
3. The input 50Ω resistor and R_L shall be type RER70 or equivalent.
4. The meter for e_i and e_o shall have a minimum bandwidth from 10 Hz to 10 kHz and shall measure true rms voltages.

FIGURE 14. Ripple rejection test circuit.

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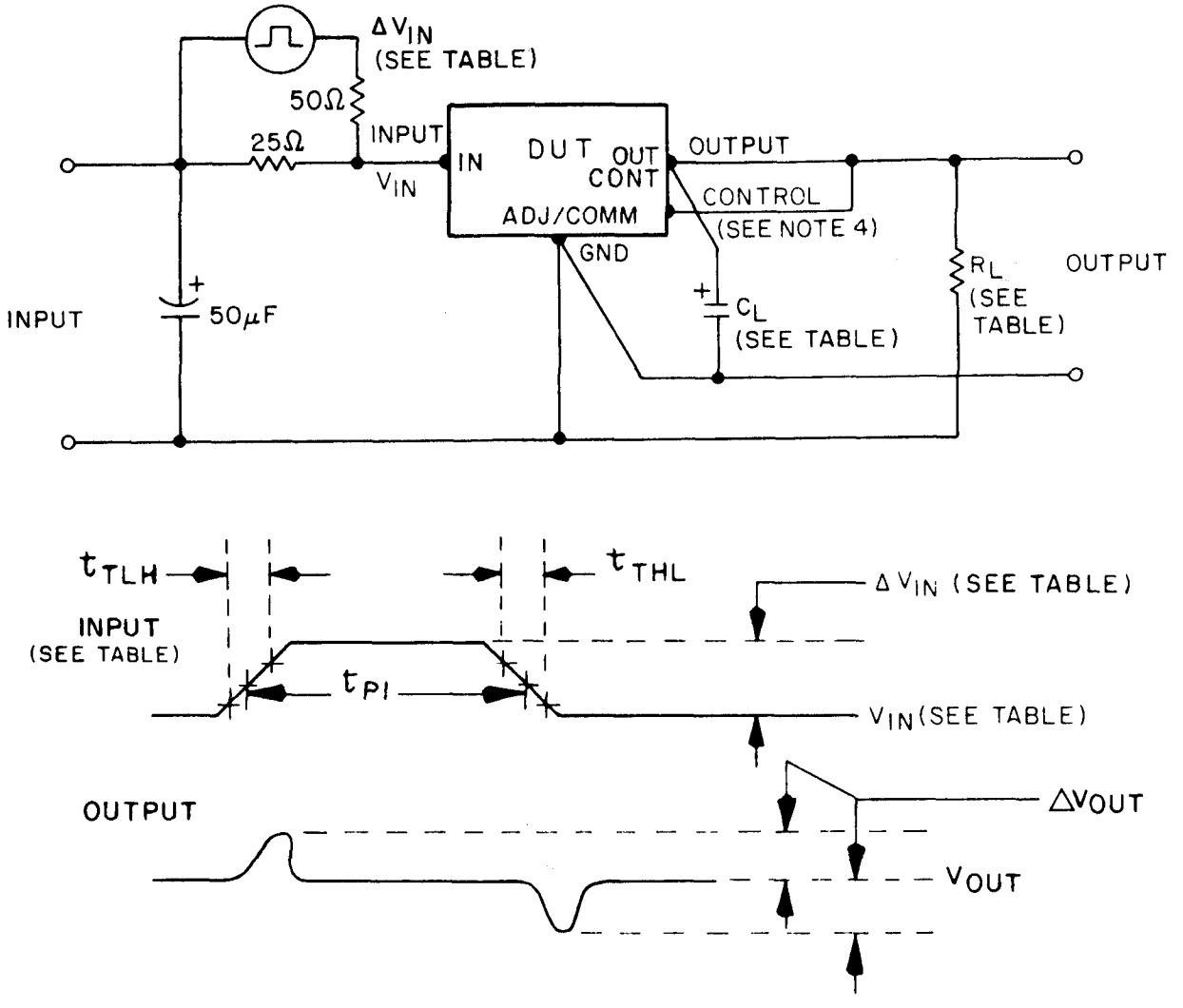
Device table

Device type	01	02	03	04	05	06
V_{IN}	10 V	10 V	6.25 V	6.25 V	6.25 V	6.25 V
R_L	100Ω	50Ω	25Ω	12.5Ω	12.5Ω	12.5Ω
C_i	0.33 µF	0.33 µF	1.0 µF	1.0 µF	1.0 µF	1.0 µF
C_L	0.1 µF	0.1 µF	1.0 µF	1.0 µF	1.0 µF	1.0 µF

NOTES:
1. The meter for measuring $e_{o\ rms}$ shall have a minimum bandwidth from 10 Hz to 10 kHz and shall measure true rms voltages.
2. $V_{NO} = e_{o\ rms}$.
3. The control pin connection is required for device types 01 and 02 only.
4. R_L shall be type RER70 or equivalent.

FIGURE 15. Noise test circuit.

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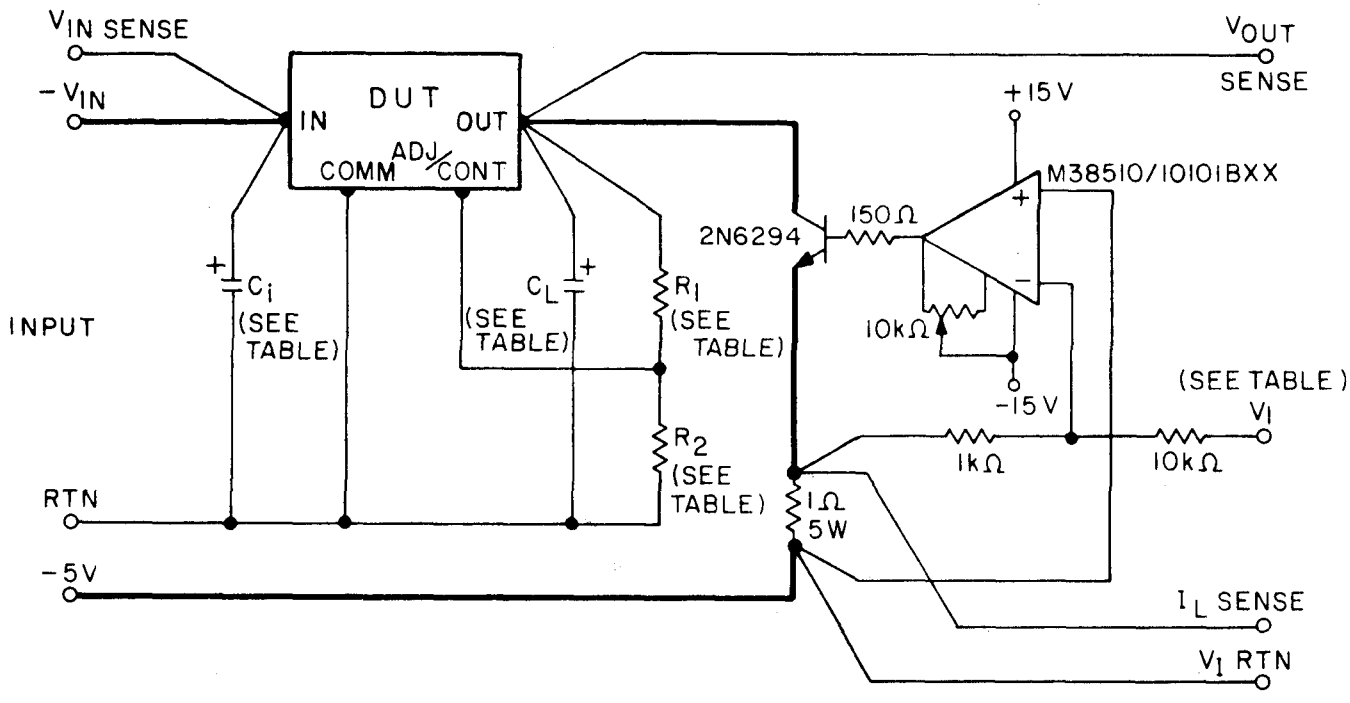
Device table

Device type	01	02	03	04	05	06	Notes
V_{IN}	10 V	10 V	6.25 V	6.25 V	6.25 V	6.25 V	1
ΔV_{IN}	3.0 V	3.0 V	3.0 V	3.0 V	3.0 V	3.0 V	1
R_L	1.25 KΩ	1.25 KΩ	120Ω	120Ω	120Ω	120Ω	
$t_{THL} = t_{TLH}$	5.0 μs	5.0 μs	5.0 μs	5.0 μs	5.0 μs	5.0 μs	1
C_L	0.1 μF	0.1 μF	1.0 μF	1.0 μF	1.0 μF	1.0 μF	

NOTES:
1. Measured at device input.
2. Pulse width t_{pl} = 25 μs; duty cycle = 3% (maximum).
3. Oscilloscope bandwidth = 5 MHz to 15 MHz.
4. The control pin connection is required for device types 01 and 02 only.
5. The input 25Ω resistor and R_L shall be type RER70 or equivalent.

FIGURE 16. Line transient response test circuit.

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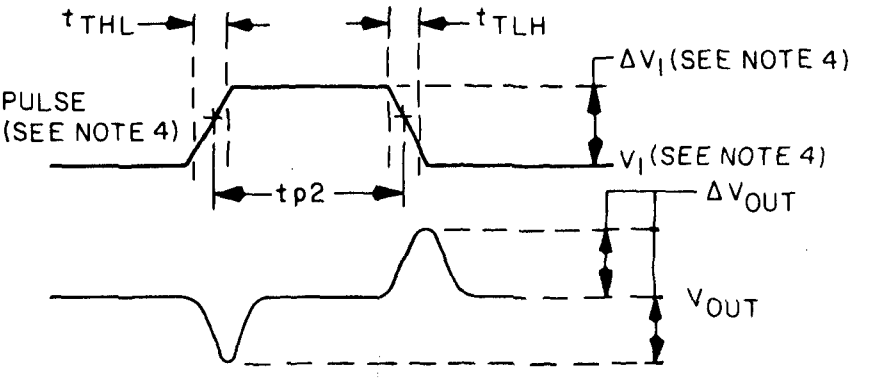


Device table

Device type	01	02	03	04	05	06
R_1	0	0	249 Ω	249 Ω	249 Ω	249 Ω
R_2	1.0 KΩ	1.0 KΩ	0	0	0	0
I_L	-50 mA	-100 mA	-50 mA	-100 mA	-100 mA	-100 mA
ΔI_L	-200 mA	-400 mA	-200 mA	-400 mA	-400 mA	-400 mA
V_I	-0.45 V	-0.95 V	-0.45 V	-0.95 V	-0.95 V	-0.95 V
ΔV_I	-2.0 V	-4.0 V	-2.0 V	-4.0 V	-4.0 V	-4.0 V
C_i	0.33 µF	0.33 µF	1.0 µF	1.0 µF	1.0 µF	1.0 µF
C_L	0.1 µF	0.1 µF	1.0 µF	1.0 µF	1.0 µF	1.0 µF

FIGURE 17. Load transient response test circuit.

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NOTES:

1. Heavy current paths ($I \geq 1.0$ A) are indicated by bold lines.
2. Kelvin connections must be used for all output current and voltage measurements.
3. The operational amplifier stabilization networks may vary with test adapter construction. Alternate drive circuits for the 2N6294 may be used to develop the proper load current and input voltage pulses. These circuits shall require the approval of the qualifying activity.
4. The pulse generator for the pulse load circuit shall have the following characteristics. (See device table III.)
   a. Voltage level (V_I) = $-10\left[|I_L| - \frac{V_{OUT}}{R_1+R_2}\right]$ volts. (Referenced to -5 Volts)
   b. Pulse width (t_{p2}) = 25 µsec.
   c. Duty cycle = 3% (maximum).
   d. $t_{THL} = t_{TLH}$ = 1.0 µsec for device types 01 and 02.
   e. $t_{THL} = t_{TLH}$ = 5.0 µsec for device types 03, 04, 05, and 06.
   f. Difference voltage level (ΔV_I) = 10 (I_L) volts.
5. a. ΔV_{OUT} = 500 mV maximum for device type 01.
   b. ΔV_{OUT} = 1,000 mV maximum for device type 02.
   c. ΔV_{OUT} = 120 mV maximum for device types 03, 04, 05, and 06.
   (These values guarantee the specified limits for load transient response.)
6. The oscilloscope shall have a bandwidth between 5 and 15 MHz.
7. Resistors R_1 and R_2 shall be type RER70 or equivalent.

FIGURE 17. Load transient response test circuit - Continued.

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TABLE III. Group A inspection for device type 01.

Subgroup	Symbol	Test no.	See figure 12 — Test conditions V_{IN} (volts)	Test conditions I_L (mA)	Applied test voltages (volts) (Hi-Lo pin potential) 1-2	4-5	6-11	7-2	8-2	Relays energized	Measurement sense lines — Pins	Value	Unit	Equation	Notes	Limits Min	Limits Max	Unit
1 (T_C = 25°C)	V_{OUT1}	1	8	-5	8	-	~	-	-	None	9-11	E_1	V	$V_{OUT1} = E_1$		4.75	5.25	V
	V_{OUT2}	2	8	-500	8	-4.95	~	-	-			E_2		$V_{OUT2} = E_2$				
	V_{OUT3}	3	30	-5	30	-	~	-	-			E_3		$V_{OUT3} = E_3$				
	V_{OUT4}	4	30	-50	30	-0.45	~	-	-			E_4		$V_{OUT4} = E_4$				
	V_{RLINE1}	5	8	-50	8	-0.45	~	-	-			E_5		$V_{RLINE1} = E_5 - E_4$	See figure 12 waveforms	-150	150	mV
	V_{RLINE2}	6	8	-350	8	-3.45	~	-	-			E_6				-	-	
	V_{RLINE2}	7	25	-350	25	-3.45	~	-	-			E_7		$V_{RLINE2} = E_6 - E_7$		-50	50	
	V_{RLOAD1}	8	10	-5	10	-	~	-	-			E_8			See figure 12 waveforms	-	-	
	V_{RLOAD1}	9	10	-500	10	-4.95	~	-	-			E_9		$V_{RLOAD1} = E_8 - E_9$		-100	100	
	V_{RLOAD2}	10	-	-	~	-	~	-	-		-	-		$V_{RLOAD2} = E_3 - E_4$		-150	150	
	V_{RTH}	11	15	-500	15	-4.95	~	-	-		9-11	E_{10}		$V_{RTH} = E_{10}$	See figure 12 waveforms t = 10.5ms	-50	50	
	I_{SCD1}	12	10	-5	10	-	~	-	-		12-13	E_{11}		$I_{SCD1} = \frac{E_{11}}{2000}$		-7.0	-0.5	mA
	I_{SCD2}	13	30	-5	30	-	~	-	-			E_{12}		$I_{SCD2} = \frac{E_{12}}{2000}$		-8.0	-0.5	
	ΔI_{SCD} (LINE)	14	8	-5	8	-	~	-	-			E_{13}		$I_{SCD(LINE)} = \frac{E_{13} - E_{12}}{2000}$		-1.0	1.0	
	ΔI_{SCD} (LOAD)	15	10	-500	10	-4.95	~	-	-			E_{14}		$I_{SCD(LOAD)} = \frac{E_{11} - E_{14}}{2000}$		-0.5	0.5	
	I_{OS1}	16	10	~	15	-	~	-1.0	0	K4,K5	10-5	E_{15}		$I_{OS1} = E_{15}$	See figure 12 waveforms	-2.00	-0.50	A
	V_{OUT5} (RECOV)	17	10	~	15	-	~	-1.0	15	K4,K5	9-11	E_{16}		$V_{OUT5} = E_{16}$		4.75	5.25	V
	I_{OS2}	18	30	~	30	-	~	-	0	K5	10-5	E_{17}		$I_{OS2} = E_{17}$		-1.00	-0.01	A
	V_{OUT6} (RECOV)	19	30	~	30	-	~	-	15	K5	9-11	E_{18}		$V_{OUT6} = E_{18}$		4.75	5.25	V
	I_{CTL}	20	10	-350	10	-3.45	~	-	-	K1,K2	12-13	E_{19}		$I_{CTL} = \frac{E_{19}}{33200}$		-5.00	-0.01	µA
	V_{START}	21	8	-500	15	-	~	-0.8	-	K4	9-11	E_{20}		$V_{OUT} = E_{20}$	See figure 12 waveforms	4.75	5.25	V
	V_{OUT7}	22	38	-500	38	-4.95	~	-	-	None		E_{21}		$V_{OUT7} = E_{21}$	R_1 = 4.99kΩ ±0.1%	28.5	31.5	
2 (T_C = 125°C)	V_{OUT1}	23	8	-5	8	-	~	-	-			E_{22}		$V_{OUT1} = E_{22}$		4.75	5.25	
	V_{OUT2}	24	8	-500	8	-4.95	~	-	-			E_{23}		$V_{OUT2} = E_{23}$				
	V_{OUT3}	25	30	-5	30	-	~	-	-			E_{24}		$V_{OUT3} = E_{24}$				
	V_{OUT4}	26	30	-50	30	-0.45	~	-	-			E_{25}		$V_{OUT4} = E_{25}$				
	V_{RLINE1}	27	8	-50	8	-0.45	~	-	-			E_{26}		$V_{RLINE1} = E_{26} - E_{25}$	See figure 12 waveforms	-150	150	mV
	V_{RLINE2}	28	8	-350	8	-3.45	~	-	-			E_{27}				-	-	
	V_{RLINE2}	29	25	-350	25	-3.45	~	-	-			E_{28}		$V_{RLINE2} = E_{27} - E_{28}$		-50	50	
	V_{RLOAD1}	30	10	-5	10	-	~	-	-			E_{29}			See figure 12 waveforms	-	-	
	V_{RLOAD1}	31	10	-500	10	-4.95	~	-	-			E_{30}		$V_{RLOAD1} = E_{29} - E_{30}$		-100	100	
	V_{RLOAD2}	32	-	-	~	-	~	-	-		-	-		$V_{RLOAD2} = E_{24} - E_{25}$		-150	150	
	I_{SCD1}	33	10	-5	10	-					12-13	E_{31}		$I_{SCD1} = \frac{E_{31}}{2000}$		-7.0	-0.5	mA
	I_{SCD2}	34	30	-5	30	-	~	-	-		12-13	E_{32}		$I_{SCD2} = \frac{E_{32}}{2000}$		-8.0	-0.5	mA

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TABLE III. Group A inspection for device type 01 - Continued.

Subgroup	Symbol	Test no.	See figure 12 Test conditions V_{IN} (volts)	I_L (mA)	Applied test voltages (volts) (Hi-Lo pin potential) 1-2	4-5	6-11	7-2	8-2	Relays energized	Measurement sense lines Pins	Value	Unit	Equation	Notes	Limits Min	Max	Unit
2 (T_C = 125°C)	ΔI_{SCD} (LINE)	35	8	-5	8	-	-	-	-	None	12-13	E_{33}	V	ΔI_{SCD} (LINE) = $\frac{E_{33}-E_{32}}{2000}$		-1.0	1.0	mA
	ΔI_{SCD} (LOAD)	36	10	-500	10	-4.95	-	-	-	None	12-13	E_{34}		ΔI_{SCD} (LOAD) = $\frac{E_{31}-E_{34}}{2000}$		-0.5	0.5	mA
	I_{OS1}	37	10	-	15	-	-	-1.0	0	K4,K5	10-5	E_{35}		$I_{OS1} = E_{35}$	See figure 12 waveforms	-2.00	-0.50	A
	V_{OUT5} (RECOV)	38	10	-	15	-	-	-1.0	15	K4,K5	9-11	E_{36}		$V_{OUT5} = E_{36}$		4.75	5.25	V
	I_{OS2}	39	30	-	30	-	-		0	K5	10-5	E_{37}		$I_{OS2} = E_{37}$		-1.00	-0.01	A
	V_{OUT6} (RECOV)	40	30	-	30	-	-		15	K5	9-11	E_{38}		$V_{OUT6} = E_{38}$		4.75	5.25	V
	I_{CTL}	41	10	-350	10	-3.45	-	-	-	K1,K2	12-13	E_{39}		$I_{CTL} = \frac{E_{39}}{33200}$		-8.00	-0.01	µA
	V_{START}	42	8	-500	15	-	-	-0.8	-	K4	9-11	E_{40}		$V_{OUT} = E_{40}$	See figure 12 waveforms	4.75	5.25	V
	V_{OUT7}	43	38	-500	38	-4.95	-	-	-	None		E_{41}		$V_{OUT7} = E_{41}$	R_1 = 4.99kΩ ±0.1%	28.5	31.5	
(T_C = 150°C)	V_{OUT8}	44	10	-5	10	-	-	-	-			E_{42}		$V_{OUT8} = E_{42}$		4.70	5.30	
3 (T_C = -55°C)	V_{OUT1}	45	8	-5	8	-	-	-	-			E_{43}		$V_{OUT1} = E_{43}$		4.75	5.25	
	V_{OUT2}	46	8	-500	8	-4.95	-	-	-			E_{44}		$V_{OUT2} = E_{44}$				
	V_{OUT3}	47	30	-5	30	-	-	-	-			E_{45}		$V_{OUT3} = E_{45}$				
	V_{OUT4}	48	30	-50	30	-0.45	-	-	-			E_{46}		$V_{OUT4} = E_{46}$				
	V_{RLINE1}	49	8	-50	8	-0.45	-	-	-			E_{47}		$V_{RLINE1} = E_{47}-E_{46}$	See figure 12 waveforms	-150	150	mV
	V_{RLINE2}	50	8	-350	8	-3.45	-	-	-			E_{48}				-	-	
	V_{RLINE2}	51	25	-350	25	-3.45	-	-	-			E_{49}		$V_{RLINE2} = E_{48}-E_{49}$		-50	50	
	V_{RLOAD1}	52	10	-5	10	-	-	-	-			E_{50}			See figure 12 waveforms	-	-	
	V_{RLOAD1}	53	10	-500	10	-4.95	-	-	-			E_{51}		$V_{RLOAD1} = E_{50}-E_{51}$		-100	100	
	V_{RLOAD2}	54	-	-	-	-	-	-	-		-	-		$V_{RLOAD2} = E_{45}-E_{46}$		-150	150	
	I_{SCD1}	55	10	-5	10	-	-	-	-		12-13	E_{52}		$I_{SCD1} = \frac{E_{52}}{2000}$		-7.0	-0.5	mA
	I_{SCD2}	56	30	-5	30	-	-	-	-			E_{53}		$I_{SCD2} = \frac{E_{53}}{2000}$		-8.0	-0.5	
	ΔI_{SCD} (LINE)	57	8	-5	8	-	-	-	-			E_{54}		ΔI_{SCD} (LINE) = $\frac{E_{54}-E_{53}}{2000}$		-1.0	1.0	
	ΔI_{SCD} (LOAD)	58	10	-500	10	-4.95	-	-	-			E_{55}		ΔI_{SCD} (LOAD) = $\frac{E_{52}-E_{55}}{2000}$		-0.5	0.5	
	I_{OS1}	59	10	-	15	-	-	-1.0	0	K4,K5	10-5	E_{56}		$I_{OS1} = E_{56}$	See figure 12 waveforms	-2.00	-0.50	A
	V_{OUT5} (RECOV)	60	10	-	15	-	-	-1.0	15	K4,K5	9-11	E_{57}		$V_{OUT5} = E_{57}$		4.75	5.25	V
	I_{OS2}	61	30	-	30	-	-	-	0	K5	10-5	E_{58}		$I_{OS2} = E_{58}$		-1.00	-0.01	A
	V_{OUT6} (RECOV)	62	30	-	30	-	-	-	15	K5	9-11	E_{59}		$V_{OUT6} = E_{59}$		4.75	5.25	V
	I_{CTL}	63	10	-350	10	-3.45	-	-	-	K1,K2	12-13	E_{60}		$I_{CTL} = \frac{E_{60}}{33200}$		-8.00	-0.01	µA

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TABLE III. Group A inspection for device type 01 - Continued.

Subgroup	Symbol	Test no.	See figure 12 Test conditions V_{IN} (volts)	I_L (mA)	Applied test voltages (volts) (Hi-Lo pin potential) 1-2	4-5	6-11	7-2	8-2	Relays energized	Measurement sense lines Pins	Value	Unit	Equation	Notes	Limits Min	Max	Unit
3 (T_C = -55°C)	V_{START}	64	8	-500	15	-	-	-0.8	-	K4	9-11	E_{61}	V	$V_{OUT} = E_{61}$	See figure 12 waveforms	4.75	5.25	V
	V_{OUT7}	65	38	-500	38	-4.95	-	-	-	None	9-11	E_{62}	V	$V_{OUT7} = E_{62}$	R_1 = 4.99 kΩ ±0.1%	28.5	31.5	V

Subgroup	Symbol	Test no.	Test conditions Input voltage	Load current	Measurement sense lines Symbol	Value	Unit	Equation	Notes	Limits Min	Max	Unit
4 (T_C = 25°C)	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	66	V_{IN} = 10V, e_i = 1.0 Vrms, @ 2400 Hz	I_L = -125mA	e_{orms}	E_{63}	Vrms	$\frac{\Delta V_{IN}}{\Delta V_{OUT}} = -20\log E_{63}$	See figure 14	45	-	dB
7 (T_C = 25°C)	V_{NO}	67	V_{IN} = 10V	I_L = -50mA	e_{orms}	E_{64}	Vrms	$V_{NO} = E_{64}$	See figure 15	-	125	μVrms
	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	68	V_{IN} = 10V, ΔV_{IN} = 3.0V	I_l = -5mA	V_{OUT}	E_{65}	V	$\frac{\Delta V_{OUT}}{\Delta V_{IN}} = \frac{E_{65}}{3}$	See figure 16	-	30	mV/V
	$\frac{\Delta V_{OUT}}{\Delta I_L}$	69	V_{IN} = 10V	I_L = -50mA, ΔI_L = -200mA	V_{OUT}	E_{66}	V	$\frac{\Delta V_{OUT}}{\Delta I_L} = \frac{E_{66}}{200}$	See figure 17	-	2.5	mV/mA

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TABLE III. Group A inspection for device type 02 - Continued.

Subgroup	Symbol	Test no.	See figure 12							Relays energized	Measurement sense lines			Equation	Notes	Limits		Unit
			Test conditions		Applied test voltages (volts) (Hi-Lo pin potential)						Pins	Value	Unit			Min	Max	
			V_{IN} (volts)	I_L (mA)	1-2	4-5	6-11	7-2	8-2									
1 (T_C = 25°C)	V_{OUT1}	1	8	-5	8	-	-	-	-	None	9-11	E_1	V	$V_{OUT1} = E_1$		4.75	5.25	V
	V_{OUT2}	2	8	-1000	8	-9.95	-	-	-			E_2		$V_{OUT2} = E_2$				
	V_{OUT3}	3	30	-5	30	-	-	-	-			E_3		$V_{OUT3} = E_3$				
	V_{OUT4}	4	30	-100	30	-0.95	-	-	-			E_4		$V_{OUT4} = E_4$				
	V_{RLINE1}	5	8	-100	8	-0.95	-	-	-			E_5		$V_{RLINE1} = E_5 - E_4$	See figure 12 waveforms	-150	150	mV
	V_{RLINE2}	6	8	-500	8	-4.95	-	-	-			E_6				-	-	
	V_{RLINE2}	7	25	-500	25	-4.95	-	-	-			E_7		$V_{RLINE2} = E_6 - E_7$		-50	50	
	V_{RLOAD1}	8	10	-5	10	-	-	-	-			E_8			See figure 12 waveforms	-	-	
	V_{RLOAD1}	9	10	-1000	10	-9.95	-	-	-			E_9		$V_{RLOAD1} = E_8 - E_9$		-100	100	
	V_{RLOAD2}	10	-	-	-	-	-	-	-		-	-		$V_{RLOAD2} = E_3 - E_4$		-150	150	
	V_{RTH}	11	15	-1000	15	-9.95	-	-	-		9-11	E_{10}		$V_{RTH} = E_{10}$	See figure 12 waveforms t = 10.5ms	-50	50	
	I_{SCD1}	12	10	-5	10	-	-	-	-		12-13	E_{11}		$I_{SCD1} = \frac{E_{11}}{2000}$		-7.0	-0.5	mA
	I_{SCD2}	13	30	-5	30	-	-	-	-			E_{12}		$I_{SCD2} = \frac{E_{12}}{2000}$		-8.0	-0.5	
	ΔI_{SCD} (LINE)	14	8	-5	8	-	-	-	-			E_{13}		ΔI_{SCD} (LINE) $= \frac{E_{13} - E_{12}}{2000}$		-1.0	1.0	
	ΔI_{SCD} (LOAD)	15	10	-1000	10	-9.95	-	-	-			E_{14}		ΔI_{SCD} (LOAD) $= \frac{E_{11} - E_{14}}{2000}$		-0.5	0.5	
	I_{OS1}	16	10	-	15	-	-	-1.0	0	K4,K5	10-5	E_{15}		$I_{OS1} = E_{15}$	See figure 12 waveforms	-4.00	-1.00	A
	V_{OUT5} (RECOV)	17	10	-	15	-	-	-1.0	15	K4,K5	9-11	E_{16}		$V_{OUT5} = E_{16}$		4.75	5.25	V
	I_{OS2}	18	30	-	30	-	-	-0	0	K5	10-5	E_{17}		$I_{OS2} = E_{17}$		-2.00	-0.02	A
	V_{OUT6} (RECOV)	19	30	-	30	-	-	-0	15	K5	9-11	E_{18}		$V_{OUT6} = E_{18}$		4.75	5.25	V
	I_{CTL}	20	10	-500	10	-4.95	-	-	-	K1,K2	12-13	E_{19}		$I_{CTL} = \frac{E_{19}}{33200}$		-5.00	-0.01	µA
	V_{START}	21	8	-1000	15	-	-	-0.8	-	K4	9-11	E_{20}		$V_{OUT} = E_{20}$	See figure 12 waveforms	4.75	5.25	V
	V_{OUT7}	22	38	-1000	38	-	-	-	-	None		E_{21}		$V_{OUT7} = E_{21}$	R_1 = 4.99kΩ ±0.1%	28.5	31.5	
2 (T_C = 125°C)	V_{OUT1}	23	8	-5	8	-	-	-	-			E_{22}		$V_{OUT1} = E_{22}$		4.75	5.25	
	V_{OUT2}	24	8	-1000	8	-9.95	-	-	-			E_{23}		$V_{OUT2} = E_{23}$				
	V_{OUT3}	25	30	-5	30	-	-	-	-			E_{24}		$V_{OUT3} = E_{24}$				
	V_{OUT4}	26	30	-100	30	-0.95	-	-	-			E_{25}		$V_{OUT4} = E_{25}$				
	V_{RLINE1}	27	8	-100	8	-0.95	-	-	-			E_{26}		$V_{RLINE1} = E_{26} - E_{25}$	See figure 12 waveforms	-150	150	mV
	V_{RLINE2}	28	8	-500	8	-4.95	-	-	-			E_{27}				-	-	
	V_{RLINE2}	29	25	-500	25	-4.95	-	-	-			E_{28}		$V_{RLINE2} = E_{27} - E_{28}$		-50	50	
	V_{RLOAD1}	30	10	-5	10	-	-	-	-			E_{29}			See figure 12 waveforms	-	-	
	V_{RLOAD1}	31	10	-1000	10	-9.95	-	-	-			E_{30}		$V_{RLOAD1} = E_{29} - E_{30}$		-100	100	
	V_{RLOAD2}	32	-	-	-	-	-	-	-		-	-		$V_{RLOAD2} = E_{24} - E_{25}$		-150	150	
	I_{SCD1}	33	10	-5	10	-					12-13	E_{31}		$I_{SCD1} = \frac{E_{31}}{2000}$		-7.0	-0.5	mA
	I_{SCD2}	34	30	-5	30	-	-	-	-		12-13	E_{32}		$I_{SCD2} = \frac{E_{32}}{2000}$		-8.0	-0.5	mA

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TABLE III. Group A inspection for device type 02 - Continued.

Subgroup	Symbol	Test no.	See figure 12 Test conditions V_{IN} (volts)	I_L (mA)	Applied test voltages (volts) (Hi-Lo pin potential) 1-2	4-5	6-11	7-2	8-2	Relays energized	Measurement sense lines Pins	Value	Unit	Equation	Notes	Limits Min	Max	Unit
2 (T_C = 125°C)	ΔI_{SCD} (LINE)	35	8	-5	8	-	-	-	-	None	12-13	E_{33}	V	ΔI_{SCD} (LINE) = $\frac{E_{33}-E_{32}}{2000}$		-1.0	1.0	mA
	ΔI_{SCD} (LOAD)	36	10	-1000	10	-9.95	-	-	-	None	12-13	E_{34}		ΔI_{SCD} (LOAD) = $\frac{E_{31}-E_{34}}{2000}$		-0.5	0.5	mA
	I_{OS1}	37	10	-	15	-	-	1.0	0	K4,K5	10-5	E_{35}		$I_{OS1} = E_{35}$	See figure 12 waveforms	-4.00	-1.00	A
	V_{OUT5} (RECOV)	38	10	-	15	-	-	1.0	15	K4,K5	9-11	E_{36}		$V_{OUT5} = E_{36}$		4.75	5.25	V
	I_{OS2}	39	30	-	30	-	-	-0	0	K5	10-5	E_{37}		$I_{OS2} = E_{37}$		-2.00	-0.02	A
	V_{OUT6} (RECOV)	40	30	-	30	-	-	-0	15	K5	9-11	E_{38}		$V_{OUT6} = E_{38}$		4.75	5.25	V
	I_{CTL}	41	10	-500	10	-4.95	-	-	-	K1,K2	12-13	E_{39}		$I_{CTL} = \frac{E_{39}}{33200}$		-8.00	-0.01	µA
	V_{START}	42	8	-1000	15	-	-	-0.8	-	K4	9-11	E_{40}		$V_{OUT} = E_{40}$	See figure 12 waveforms	4.75	5.25	V
	V_{OUT7}	43	38	-1000	38	-	-	-	-	None		E_{41}		$V_{OUT7} = E_{41}$	R_1 = 4.99kΩ ±0.1%	28.5	31.5	
(T_C = 150°C)	V_{OUT8}	44	10	-5	10	-	-	-	-			E_{42}		$V_{OUT8} = E_{42}$		4.70	5.30	
3 (T_C = -55°C)	V_{OUT1}	45	8	-5	8	-	-	-	-			E_{43}		$V_{OUT1} = E_{43}$		4.75	5.25	
	V_{OUT2}	46	8	-1000	8	-9.95	-	-	-			E_{44}		$V_{OUT2} = E_{44}$				
	V_{OUT3}	47	30	-5	30	-	-	-	-			E_{45}		$V_{OUT3} = E_{45}$				
	V_{OUT4}	48	30	-100	30	-0.95	-	-	-			E_{46}		$V_{OUT4} = E_{46}$				
	V_{RLINE1}	49	8	-100	8	-0.95	-	-	-			E_{47}		$V_{RLINE1} = E_{47}-E_{46}$	See figure 12 waveforms	-150	150	mV
	V_{RLINE2}	50	8	-500	8	-4.95	-	-	-			E_{48}				-	-	
	V_{RLINE2}	51	25	-500	25	-4.95	-	-	-			E_{49}		$V_{RLINE2} = E_{48}-E_{49}$		-50	50	
	V_{RLOAD1}	52	10	-5	10	-	-	-	-			E_{50}			See figure 12 waveforms	-	-	
	V_{RLOAD1}	53	10	-1000	10	-9.95	-	-	-			E_{51}		$V_{RLOAD1} = E_{50}-E_{51}$		-100	100	
	V_{RLOAD2}	54	-	-	-	-	-	-	-		-	-		$V_{RLOAD2} = E_{45}-E_{46}$		-150	150	
	I_{SCD1}	55	10	-5	10	-	-	-	-		12-13	E_{52}		$I_{SCD1} = \frac{E_{52}}{2000}$		-7.0	-0.5	mA
	I_{SCD2}	56	30	-5	30	-	-	-	-			E_{53}		$I_{SCD2} = \frac{E_{53}}{2000}$		-8.0	-0.5	
	ΔI_{SCD} (LINE)	57	8	-5	8	-	-	-	-			E_{54}		ΔI_{SCD} (LINE) = $\frac{E_{54}-E_{53}}{2000}$		-1.0	1.0	
	ΔI_{SCD} (LOAD)	58	10	-1000	10	-9.95	-	-	-			E_{55}		ΔI_{SCD} (LOAD) = $\frac{E_{52}-E_{55}}{2000}$		-0.5	0.5	
	I_{OS1}	59	10	-	15	-	-	-1.0	0	K4,K5	10-5	E_{56}		$I_{OS1} = E_{56}$	See figure 12 waveforms	-4.00	-1.00	A
	V_{OUT5} (RECOV)	60	10	-	15	-	-	-1.0	15	K4,K5	9-11	E_{57}		$V_{OUT5} = E_{57}$		4.75	5.25	V
	I_{OS2}	61	30	-	30	-	-	-	0	K5	10-5	E_{58}		$I_{OS2} = E_{58}$		-2.00	-0.02	A
	V_{OUT6} (RECOV)	62	30	-	30	-	-	-	15	K5	9-11	E_{59}		$V_{OUT6} = E_{59}$		4.75	5.25	V
	I_{CTL}	63	10	-500	10	-4.95	-	-	-	K1,K2	12-13	E_{60}		$I_{CTL} = \frac{E_{60}}{33200}$		-8.00	-0.01	µA

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TABLE III. Group A inspection for device type 02 - Continued.

Subgroup	Symbol	Test no.	See figure 12							Relays energized	Measurement sense lines			Equation	Notes	Limits		Unit
			Test conditions		Applied test voltages (Hi-Lo pin potential)													
			V_{IN} (volts)	I_L (mA)	1-2	4-5	6-11	7-2	8-2		Pins	Value	Unit			Min	Max	
3 (T_C = -55°C)	V_{START}	64	8	-1000	15	-	-	-0.8	-	K4	9-11	E_{61}	V	$V_{OUT} = E_{61}$	See figure 12 waveforms	4.75	5.25	V
	V_{OUT7}	65	38	-1000	38	-	-	-	-	None	9-11	E_{62}	V	$V_{OUT7} = E_{62}$	R_1 = 4.99 kΩ ± 0.1%	28.5	31.5	V

Subgroup	Symbol	Test no.	Test conditions		Measurement sense lines			Equation	Notes	Limits		Unit
			Input voltage	Load current	Symbol	Value	Unit			Min	Max	
4 (T_C = 25°C)	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	66	V_{IN} = 10V, e_i = 1.0Vrms @ 2400 Hz	I_L = -350mA	e_{orms}	E_{63}	Vrms	$\frac{\Delta V_{IN}}{\Delta V_{OUT}} = -20\log E_{63}$	See figure 14	45	-	dB
7 (T_C = 25°C)	V_{NO}	67	V_{IN} = 10V	I_L = -100mA	e_{orms}	E_{64}	Vrms	$V_{NO} = E_{64}$	See figure 15	-	250	μVrms
	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	68	V_{IN} = 10V, ΔV_{IN} = 3.0V	I_L = -5mA	V_{OUT}	E_{65}	V	$\frac{\Delta V_{OUT}}{\Delta V_{IN}} = \frac{E_{65}}{3}$	See figure 16	-	30	mV/V
	$\frac{\Delta V_{OUT}}{\Delta I_L}$	69	V_{IN} = 10V	I_L = -100mA, ΔI_L = -400mA	V_{OUT}	E_{66}	V	$\frac{\Delta V_{OUT}}{\Delta I_L} = \frac{E_{66}}{400}$	See figure 17	-	2.5	mV/mA

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TABLE III. Group A inspection for device type 03.

Subgroup	Symbol	Test no.	See figure 12: Test conditions V_{IN} (volts)	I_L (mA)	Applied test voltages (volts) (Hi-Lo pin potential) 1-2	4-5	6-11	7-2	8-2	Relays energized	Measurement sense lines: Pins	Value	Unit	Equation	Notes	Limits: Min	Max	Unit
1 (T_C = 25°C)	V_{OUT1}	1	4.25	-5	4.25	0	-	-	-	None	9-11	E_1	V	$V_{OUT1} = E_1$		1.20	1.30	V
	V_{OUT2}	2	4.25	-500	4.25	-4.95	-	-	-			E_2		$V_{OUT2} = E_2$				
	V_{OUT3}	3	41.25	-5	41.25	0	-	-	-			E_3		$V_{OUT3} = E_3$				
	V_{OUT4}	4	41.25	-50	41.25	-0.45	-	-	-			E_4		$V_{OUT4} = E_4$				
	V_{RLINE}	5	-	-	-	-	-	-	-			-		$V_{RLINE} = E_1 - E_3$	See figure 12 waveforms	-9	9	mV
	V_{RLOAD1}	6	6.25	-5	6.25	0	-	-	-			E_5		$V_{RLOAD1} = E_5 - E_6$	See figure 12 waveforms	-3.5	3.5	
	V_{RLOAD1}	7	6.25	-500	6.25	-4.95	-	-	-			E_6						
	V_{RLOAD2}	8	-	-	-	-	-	-	-			-		$V_{RLOAD2} = E_3 - E_4$		-3.5	3.5	
	V_{RTH}	9	14.6	-750	14.6	-7.45	-	-	-			E_7		$V_{RTH} = E_7$	See figure 12 waveforms t = 20.5ms	-5	5	
	I_{ADJ}	10	4.25	-5	4.25	0	-	-	-	K2	12-13	E_8	mV	$I_{ADJ} = \frac{E_8}{2000}$		-100	-15	µA
	I_{ADJ}	11	41.25	-5	41.25	0	-	-	-			E_9		$I_{ADJ} = \frac{E_9}{2000}$		-100	-15	
	ΔI_{ADJ} (LINE)	12	-	-	-	-	-	-	-			-		ΔI_{ADJ} (LINE) $= \frac{E_8 - E_9}{2000}$		-5	5	
	ΔI_{ADJ} (LOAD)	13	6.25	-5	6.25	0	-	-	-			E_{10}		ΔI_{ADJ} (LOAD) $= \frac{E_{10} - E_{11}}{2000}$		-5	5	
	ΔI_{ADJ} (LOAD)	14	6.25	-500	6.25	-4.95	-	-	-			E_{11}						
	I_{OS1}	15	4.25	-	10	-	-	-0.425	0	K4,K5	10-5	E_{12}	V	$I_{OS1} = E_{12}$	See figure 12 waveforms	-1.8	-0.5	A
	V_{OUT5} (RECOV)	16	4.25	-	10	-	-	-0.425	15	K4,K5	9-11	E_{13}		V_{OUT5} (RECOV) $= E_{13}$		1.20	1.30	V
	I_{OS2}	17	40	-	40	-	-	-	0	K5	10-5	E_{14}		$I_{OS2} = E_{14}$		-0.5	-0.05	A
	V_{OUT6} (RECOV)	18	40	-	40	-	-	-	15	K5	9-11	E_{15}		V_{OUT6} (RECOV) $= E_{15}$		1.20	1.30	V
	I_{Q1}	19	4.25	-	4.25	0	1.4	-	-	K3	12-13	E_{16}		$I_{Q1} = \frac{E_{16}}{2000}$		-3.0	-0.5	mA
	I_{Q2}	20	14.25	-	14.25	0	1.4	-	-	K3	12-13	E_{17}		$I_{Q2} = \frac{E_{17}}{2000}$		-3.0	-0.5	mA
	I_{Q3}	21	41.25	-	41.25	0	1.4	-	-	K3	12-13	E_{18}		$I_{Q3} = \frac{E_{18}}{2000}$		-5.0	-1.0	mA
	V_{START}	22	4.25	-500	10	0	-	-0.425	-	K4	9-11	E_{19}		$V_{START} = E_{19}$	See figure 12 waveforms	1.20	1.30	V
2 (T_C = 125°C)	V_{OUT1}	23	4.25	-5	4.25	0	-	-	-	None		E_{20}		$V_{OUT1} = E_{20}$				
	V_{OUT2}	24	4.25	-500	4.25	-4.95	-	-	-			E_{21}		$V_{OUT2} = E_{21}$				
	V_{OUT3}	25	41.25	-5	41.25	0	-	-	-			E_{22}		$V_{OUT3} = E_{22}$				
	V_{OUT4}	26	41.25	-50	41.25	-0.45	-	-	-			E_{23}		$V_{OUT4} = E_{23}$				
	V_{RLINE}	27	-	-	-	-	-	-	-			-		$V_{RLINE} = E_{20} - E_{22}$	See figure 12 waveforms	-23	23	mV
	V_{RLOAD1}	28	6.25	-5	6.25	0	-	-	-			E_{24}		$V_{RLOAD1} = E_{24} - E_{25}$	See figure 12 waveforms	-12	12	
	V_{RLOAD1}	29	6.25	-500	6.25	-4.95	-	-	-			E_{25}						
	V_{RLOAD2}	30	-	-	-	-	-	-	-			-		$V_{RLOAD2} = E_{22} - E_{23}$		-12	12	

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TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	Test no.	See figure 12 Test conditions V_{IN} (volts)	I_L (mA)	Applied test voltages (volts) (Hi-Lo pin potential) 1-2	4-5	6-11	7-2	8-2	Relays energized	Measurement sense lines Pins	Value	Unit	Equation	Notes	Limits Min	Max	Unit
2 (T_C = 125°C)	I_{ADJ}	31	4.25	-5	4.25	0	-	-	-	K2	12-13	E_{26}	mV	$I_{ADJ} = \frac{E_{26}}{2000}$		-100	-15	µA
	I_{ADJ}	32	41.25	-5	41.25	0	-	-	-			E_{27}		$I_{ADJ} = \frac{E_{27}}{2000}$		-100	-15	
	ΔI_{ADJ} (LINE)	33	-	-	-	-	-	-	-			-		ΔI_{ADJ} (LINE) $= \frac{E_{26}-E_{27}}{2000}$		-5	5	
	ΔI_{ADJ} (LOAD)	34	6.25	-5	6.25	0	-	-	-			E_{28}		ΔI_{ADJ} (LOAD) $= \frac{E_{28}-E_{29}}{2000}$		-5	5	
	ΔI_{ADJ} (LOAD)	35	6.25	-500	6.25	-4.95	-	-	-			E_{29}						
	I_{OS1}	36	4.25	-	10	-	-	-0.425	0	K4,K5	10-5	E_{30}	V	$I_{OS1} = E_{30}$	See figure 12 waveforms	-1.8	-0.5	A
	V_{OUT5} (RECOV)	37	4.25	-	10	-	-	-0.425	15	K4,K5	9-11	E_{31}		V_{OUT5} (RECOV) $= E_{31}$		1.20	1.30	V
	I_{OS2}	38	40	-	40	-	-	-	0	K5	10-5	E_{32}		$I_{OS2} = E_{32}$		-0.5	-0.05	A
	V_{OUT6} (RECOV)	39	40	-	40	-	-	-	15	K5	9-11	E_{33}		V_{OUT6} (RECOV) $= E_{33}$		1.20	1.30	V
	I_{Q1}	40	4.25	-	4.25	0	1.4	-	-	K3	12-13	E_{34}		$I_{Q1} = \frac{E_{34}}{2000}$		-3.0	-0.5	mA
	I_{Q2}	41	14.25	-	14.25	0	1.4	-	-	K3	12-13	E_{35}		$I_{Q2} = \frac{E_{35}}{2000}$		-3.0	-0.5	
	I_{Q3}	42	41.25	-	41.25	0	1.4	-	-	K3	12-13	E_{36}		$I_{Q3} = \frac{E_{36}}{2000}$		-5.0	-1.0	
	V_{START}	43	4.25	-500	10	0	-	-0.425	-	K4	9-11	E_{37}		$V_{START} = E_{37}$	See figure 12 waveforms	1.20	1.30	V
(T_C = 150°C)	V_{OUT7}	44	6.25	-5	6.25	0	-	-	-	None		E_{38}		$V_{OUT7} = E_{38}$				
3 (T_C = -55°C)	V_{OUT1}	45	4.25	-5	4.25	0	-	-	-			E_{39}		$V_{OUT1} = E_{39}$				
	V_{OUT2}	46	4.25	-500	4.25	-4.95	-	-	-			E_{40}		$V_{OUT2} = E_{40}$				
	V_{OUT3}	47	41.25	-5	41.25	0	-	-	-			E_{41}		$V_{OUT3} = E_{41}$				
	V_{OUT4}	48	41.25	-50	41.25	-0.45	-	-	-			E_{42}		$V_{OUT4} = E_{42}$				
	V_{RLINE}	49	-	-	-	-	-	-	-			-		$V_{RLINE} = E_{39}-E_{41}$	See figure 12 waveforms	-23	23	mV
	V_{RLOAD1}	50	6.25	-5	6.25	0	-	-	-			E_{43}			See figure 12 waveforms			
	V_{RLOAD1}	51	6.25	-500	6.25	-4.95	-	-	-			E_{44}		$V_{RLOAD1} = E_{43}-E_{44}$		-12	12	
	V_{RLOAD2}	52	-	-	-	-	-	-	-			-		$V_{RLOAD2} = E_{41}-E_{42}$		-12	12	
	I_{ADJ}	53	4.25	-5	4.25	0	-	-	-	K2	12-13	E_{45}	mV	$I_{ADJ} = \frac{E_{45}}{2000}$		-100	-15	µA
	I_{ADJ}	54	41.25	-5	41.25	0	-	-	-			E_{46}		$I_{ADJ} = \frac{E_{46}}{2000}$		-100	-15	
	ΔI_{ADJ} (LINE)	55	-	-	-	-	-	-	-			-		ΔI_{ADJ} (LINE) $= \frac{E_{45}-E_{46}}{2000}$		-5	5	
	ΔI_{ADJ} (LOAD)	56	6.25	-5	6.25	0	-	-	-			E_{47}		ΔI_{ADJ} (LOAD) $= \frac{E_{47}-E_{48}}{2000}$		-5	5	
	ΔI_{ADJ} (LOAD)	57	6.25	-500	6.25	-4.95	-	-	-			E_{48}						

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TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	Test no.	See figure 12							Relays energized	Measurement sense lines			Equation	Notes	Limits		Unit
			Test conditions		Applied test voltages (volts) (Hi-Lo pin potential)						Pins	Value	Unit			Min	Max	
			V_{IN} (volts)	I_L (mA)	1-2	4-5	6-11	7-2	8-2									
3 (T_C = -55°C)	I_{OS1}	58	4.25	-	10	-	-	-0.425	0	K4,K5	10-5	E_{49}	V	$I_{OS1} = E_{49}$	See figure 12 waveforms	-1.8	-0.5	A
	V_{OUT5} (RECOV)	59	4.25	-	10	-	-	-0.425	15	K4,K5	9-11	E_{50}		$V_{OUT5} = E_{50}$ (RECOV)		1.20	1.30	V
	I_{OS2}	60	40	-	40	-	-	-	0	K5	10-5	E_{51}		$I_{OS2} = E_{51}$		-0.5	-0.05	A
	V_{OUT6} (RECOV)	61	40	-	40	-	-	-	15	K5	9-11	E_{52}		$V_{OUT6} = E_{52}$ (RECOV)		1.20	1.30	V
	I_{Q1}	62	4.25	-	4.25	0	1.4	-	-	K3	12-13	E_{53}		$I_{Q1} = \frac{E_{53}}{2000}$		-3.0	-0.5	mA
	I_{Q2}	63	14.25	-	14.25	0	1.4	-	-	K3	12-13	E_{54}		$I_{Q2} = \frac{E_{54}}{2000}$		-3.0	-0.5	mA
	I_{Q3}	64	41.25	-	41.25	0	1.4	-	-	K3	12-13	E_{55}		$I_{Q3} = \frac{E_{55}}{2000}$		-5.0	-1.0	mA
	V_{START}	65	4.25	-500	10	0	-	-0.425	-	K4	9-11	E_{56}		$V_{START} = E_{56}$	See figure 12 waveforms	1.20	1.30	V

Subgroup	Symbol	Test no.	Test conditions		Measurement sense lines			Equation	Notes	Limits		Unit
			Input voltage	Load current	Symbol	Value	Unit			Min	Max	
4 (T_C = 25°C)	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	66	V_{IN} = 6.25V, e_i = 1.0Vrms @ 2400 Hz	I_L = -125mA	e_{orms}	E_{57}	Vrms	$\frac{\Delta V_{IN}}{\Delta V_{OUT}} = -20\log E_{57}$	See figure 14	65	-	dB
7 (T_C = 25°C)	V_{NO}	67	V_{IN} = 6.25V	I_L = -50mA	e_{orms}	E_{58}	Vrms	$V_{NO} = E_{58}$	See figure 15	-	120	µVrms
	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	68	V_{IN} = 6.25V, ΔV_{IN} = 3.0V	I_L = -10mA	V_{OUT}	E_{59}	V	$\frac{\Delta V_{OUT}}{\Delta V_{IN}} = \frac{E_{59}}{3}$	See figure 16	-	6	mV/V
	$\frac{\Delta V_{OUT}}{\Delta I_L}$	69	V_{IN} = 6.25V	I_L = -50mA, ΔI_L = -200mA	V_{OUT}	E_{60}	V	$\frac{\Delta V_{OUT}}{\Delta I_L} = \frac{E_{60}}{200}$	See figure 17	-	0.6	mV/mA

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TABLE III. Group A inspection for device type 04.

Subgroup	Symbol	Test no.	See figure 12 — Test conditions V_{IN} (volts)	Test conditions I_L (mA)	Applied test voltages (volts) (Hi-Lo pin potential) 1-2	4-5	6-11	7-2	8-2	Relays energized	Measurement sense lines — Pins	Value	Unit	Equation	Notes	Limits Min	Limits Max	Unit
1 (T_C = 25°C)	V_{OUT1}	1	4.25	-5	4.25	0	-	-	-	None	9-11	E_1	V	$V_{OUT1} = E_1$		1.20	1.30	V
	V_{OUT2}	2	4.25	-1500	4.25	-14.95	-	-	-			E_2		$V_{OUT2} = E_2$				
	V_{OUT3}	3	41.25	-5	41.25	0	-	-	-			E_3		$V_{OUT3} = E_3$				
	V_{OUT4}	4	41.25	-200	41.25	-1.95	-	-	-			E_4		$V_{OUT4} = E_4$				
	V_{RLINE}	5	-	-	-	-	-	-	-			-		$V_{RLINE} = E_1 - E_3$	See figure 12 waveforms	-9	9	mV
	V_{RLOAD1}	6	6.25	-5	6.25	0	-	-	-			E_5		$V_{RLOAD1} = E_5 - E_6$	See figure 12 waveforms	3.5	3.5	
	V_{RLOAD1}	7	6.25	-1500	6.25	-14.95	-	-	-			E_6				-3.5	3.5	
	V_{RLOAD2}	8	-	-	-	-	-	-	-			-		$V_{RLOAD2} = E_3 - E_4$		-3.5	3.5	
	V_{RTH}	9	14.6	-1500	14.6	-14.95	-	-	-			E_7		$V_{RTH} = E_7$	See figure 12 waveforms t = 20.5ms	-5	5	
	I_{ADJ}	10	4.25	-5	4.25	0	-	-	-	K2	12-13	E_8	mV	$I_{ADJ} = \frac{E_8}{2000}$		-100	-15	µA
	I_{ADJ}	11	41.25	-5	41.25	0	-	-	-			E_9		$I_{ADJ} = \frac{E_9}{2000}$		-100	-15	
	ΔI_{ADJ} (LINE)	12	-	-	-	-	-	-	-			-		ΔI_{ADJ} (LINE) $= \frac{E_8 - E_9}{2000}$		-5	5	
	ΔI_{ADJ} (LOAD)	13	6.25	-5	6.25	0	-	-	-			E_{10}		ΔI_{ADJ} (LOAD) $= \frac{E_{10} - E_{11}}{2000}$		-5	5	
	ΔI_{ADJ} (LOAD)	14	6.25	-1500	6.25	-14.95	-	-	-			E_{11}						
	I_{OS1}	15	4.25	-	10	-	-	-0.425	0	K4,K5	10-5	E_{12}	V	$I_{OS} = E_{12}$	See figure 12 waveforms	-3.5	-1.5	A
	V_{OUT5} (RECOV)	16	4.25	-	10	-	-	-0.425	15	K4,K5	9-11	E_{13}		$V_{OUT5} = E_{13}$ (RECOV)		1.20	1.30	V
	I_{OS2}	17	40	-	40	-	-	-	0	K5	10-5	E_{14}		$I_{OS} = E_{14}$		-1.00	-0.18	A
	V_{OUT6} (RECOV)	18	40	-	40	-	-	-	15	K5	9-11	E_{15}		$V_{OUT6} = E_{15}$ (RECOV)		1.20	1.30	V
	I_{Q1}	19	4.25	-	4.25	0	1.4	-	-	K3	12-13	E_{16}		$I_{Q1} = \frac{E_{16}}{2000}$		-3.0	-0.5	mA
	I_{Q2}	20	14.25	-	14.25	0	1.4	-	-	K3	12-13	E_{17}		$I_{Q2} = \frac{E_{17}}{2000}$		-3.0	-0.5	mA
	I_{Q3}	21	41.25	-	41.25	0	1.4	-	-	K3	12-13	E_{18}		$I_{Q3} = \frac{E_{18}}{2000}$		-5.0	-1.0	mA
	V_{START}	22	4.25	-1500	10	0	-	-0.425	-	K4	9-11	E_{19}		$V_{START} = E_{19}$	See figure 12 waveforms	1.20	1.30	V
2 (T_C = 125°C)	V_{OUT1}	23	4.25	-5	4.25	0	-	-	-	None		E_{20}		$V_{OUT1} = E_{20}$				
	V_{OUT2}	24	4.25	-1500	4.25	-14.95	-	-	-			E_{21}		$V_{OUT2} = E_{21}$				
	V_{OUT3}	25	41.25	-5	41.25	0	-	-	-			E_{22}		$V_{OUT3} = E_{22}$				
	V_{OUT4}	26	41.25	-200	41.25	-1.95	-	-	-			E_{23}		$V_{OUT4} = E_{23}$				
	V_{RLINE}	27	-	-	-	-	-	-	-			-		$V_{RLINE} = E_{20} - E_{22}$	See figure 12 waveforms	-23	23	mV
	V_{RLOAD1}	28	6.25	-5	6.25	0	-	-	-			E_{24}		$V_{RLOAD1} = E_{24} - E_{25}$	See figure 12 waveforms	-12	12	
	V_{RLOAD1}	29	6.25	-1500	6.25	-14.95	-	-	-			E_{25}				-12	12	
	V_{RLOAD2}	30	-	-	-	-	-	-	-			-		$V_{RLOAD2} = E_{22} - E_{23}$				

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TABLE III. Group A inspection for device type 04 - Continued.

Subgroup	Symbol	Test no.	See figure 12 Test conditions V_{IN} (volts)	I_L (mA)	Applied test voltages (volts) (Hi-Lo pin potential) 1-2	4-5	6-11	7-2	8-2	Relays energized	Measurement sense lines Pins	Value	Unit	Equation	Notes	Limits Min	Max	Unit
2 (T_C = 125°C)	I_{ADJ}	31	4.25	-5	4.25	0	-	-	-	K2	12-13	E_{26}	mV	$I_{ADJ} = \frac{E_{26}}{2000}$		-100	-15	µA
	I_{ADJ}	32	41.25	-5	41.25	0	-	-	-			E_{27}		$I_{ADJ} = \frac{E_{27}}{2000}$		-100	-15	
	ΔI_{ADJ} (LINE)	33	-	-	-	-	-	-	-			-		ΔI_{ADJ} (LINE) $= \frac{E_{26}-E_{27}}{2000}$		-5	5	
	ΔI_{ADJ} (LOAD)	34	6.25	-5	6.25	0	-	-	-			E_{28}		ΔI_{ADJ} (LOAD) $= \frac{E_{28}-E_{29}}{2000}$		-5	5	
	ΔI_{ADJ} (LOAD)	35	6.25	-1500	6.25	-14.95	-	-	-			E_{29}						
	I_{OS1}	36	4.25	-	10	-	-	-0.425	0	K4,K5	10-5	E_{30}	V	$I_{OS1} = E_{30}$	See figure 12 waveforms	-3.5	-1.5	A
	V_{OUT5} (RECOV)	37	4.25	-	10	-	-	-0.425	15	K4,K5	9-11	E_{31}		V_{OUT5} (RECOV) $= E_{31}$		1.20	1.30	V
	I_{OS2}	38	40	-	40	-	-	-	0	K5	10-5	E_{32}		$I_{OS2} = E_{32}$		-1.00	-0.18	A
	V_{OUT6} (RECOV)	39	40	-	40	-	-	-	15	K5	9-11	E_{33}		V_{OUT6} (RECOV) $= E_{33}$		1.20	1.30	V
	I_{Q1}	40	4.25	-	4.25	0	1.4	-	-	K3	12-13	E_{34}		$I_{Q1} = \frac{E_{34}}{2000}$		-3.0	-0.5	mA
	I_{Q2}	41	14.25	-	14.25	0	1.4	-	-	K3	12-13	E_{35}		$I_{Q2} = \frac{E_{35}}{2000}$		-3.0	-0.5	mA
	I_{Q3}	42	41.25	-	41.25	0	1.4	-	-	K3	12-13	E_{36}		$I_{Q3} = \frac{E_{36}}{2000}$		-5.0	-1.0	mA
	V_{START}	43	4.25	-1500	10	0	-	-0.425	-	K4	9-11	E_{37}		$V_{START} = E_{37}$	See figure 12 waveforms	1.20	1.30	V
(T_C = 150°C)	V_{OUT7}	44	6.25	-5	6.25	0	-	-	-	None		E_{38}		$V_{OUT7} = E_{38}$				
3 (T_C = -55°C)	V_{OUT1}	45	4.25	-5	4.25	0	-	-	-			E_{39}		$V_{OUT1} = E_{39}$				
	V_{OUT2}	46	4.25	-1500	4.25	-14.95	-	-	-			E_{40}		$V_{OUT2} = E_{40}$				
	V_{OUT3}	47	41.25	-5	41.25	0	-	-	-			E_{41}		$V_{OUT3} = E_{41}$				
	V_{OUT4}	48	41.25	-200	41.25	-1.95	-	-	-			E_{42}		$V_{OUT4} = E_{42}$				
	V_{RLINE}	49	-	-	-	-	-	-	-			-		$V_{RLINE} = E_{39}-E_{41}$	See figure 12 waveforms	-23	23	mV
	V_{RLOAD1}	50	6.25	-5	6.25	0	-	-	-			E_{43}		———	See figure 12 waveforms			
	V_{RLOAD1}	51	6.25	-1500	6.25	-14.95	-	-	-			E_{44}		$V_{RLOAD1} = E_{43}-E_{44}$		-12	12	
	V_{RLOAD2}	52	-	-	-	-	-	-	-			-		$V_{RLOAD2} = E_{41}-E_{42}$		-12	12	
	I_{ADJ}	53	4.25	-5	4.25	0	-	-	-	K2	12-13	E_{45}	mV	$I_{ADJ} = \frac{E_{45}}{2000}$		-100	-15	µA
	I_{ADJ}	54	41.25	-5	41.25	0	-	-	-			E_{46}		$I_{ADJ} = \frac{E_{46}}{2000}$		-100	-15	
	ΔI_{ADJ} (LINE)	55	-	-	-	-	-	-	-			-		ΔI_{ADJ} (LINE) $= \frac{E_{45}-E_{46}}{2000}$		-5	5	
	ΔI_{ADJ} (LOAD)	56	6.25	-5	6.25	0	-	-	-			E_{47}		ΔI_{ADJ} (LOAD) $= \frac{E_{47}-E_{48}}{2000}$		-5	5	
	ΔI_{ADJ} (LOAD)	57	6.25	-1500	6.25	-14.95	-	-	-			E_{48}						

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TABLE III. Group A inspection for device type 04 - Continued.

Subgroup	Symbol	Test no.	See figure 12 — Test conditions V_{IN} (volts)	I_L (mA)	Applied test voltages (volts) (Hi-Lo pin potential) 1-2	4-5	6-11	7-2	8-2	Relays energized	Measurement sense lines Pins	Value	Unit	Equation	Notes	Limits Min	Max	Unit
3 (T_C = -55°C)	I_{OS1}	58	4.25	-	10	-	-	-0.425	0	K4,K5	10-5	E_{49}	V	$I_{OS1} = E_{49}$	See figure 12 waveforms	-3.5	-1.5	A
	V_{OUT5} (RECOV)	59	4.25	-	10	-	-	-0.425	15	K4,K5	9-11	E_{50}		$V_{OUT5} = E_{50}$ (RECOV)		1.20	1.30	V
	I_{OS2}	60	40	-	40	-	-	-	0	K5	10-5	E_{51}		$I_{OS2} = E_{51}$		-1.00	-0.18	A
	V_{OUT6} (RECOV)	61	40	-	40	-	-	-	15	K5	9-11	E_{52}		$V_{OUT6} = E_{52}$ (RECOV)		1.20	1.30	V
	I_{Q1}	62	4.25	-	4.25	0	1.4	-	-	K3	12-13	E_{53}		$I_{Q1} = \frac{E_{53}}{2000}$		-3.0	-0.5	mA
	I_{Q2}	63	14.25	-	14.25	0	1.4	-	-	K3	12-13	E_{54}		$I_{Q2} = \frac{E_{54}}{2000}$		-3.0	-0.5	mA
	I_{Q3}	64	41.25	-	41.25	0	1.4	-	-	K3	12-13	E_{55}		$I_{Q3} = \frac{E_{55}}{2000}$		-5.0	-1.0	mA
	V_{START}	65	4.25	-1500	10	0	-	-0.425	-	K4	9-11	E_{56}		$V_{START} = E_{56}$	See figure 12 waveforms	1.20	1.30	V

Subgroup	Symbol	Test no.	Test conditions — Input voltage	Load current	Measurement sense lines Symbol	Value	Unit	Equation	Notes	Limits Min	Max	Unit
4 (T_C = 25°C)	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	66	V_{IN} = 6.25V, e_i = 1.0Vrms, @ 2400 Hz	I_L = -500mA	e_{orms}	E_{57}	Vrms	$\frac{\Delta V_{IN}}{\Delta V_{OUT}} = -20\log E_{57}$	See figure 14	65	-	dB
7 (T_C = 25°C)	V_{NO}	67	V_{IN} = 6.25V	I_L = -100mA	e_{orms}	E_{58}	Vrms	$V_{NO} = E_{58}$	See figure 15	-	120	µVrms
	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	68	V_{IN} = 6.25V, ΔV_{IN} = 3.0V	I_L = -10mA	V_{OUT}	E_{59}	V	$\frac{\Delta V_{OUT}}{\Delta V_{IN}} = \frac{E_{59}}{3}$	See figure 16	-	6	mV/V
	$\frac{\Delta V_{OUT}}{\Delta I_L}$	69	V_{IN} = 6.25V	I_L = -100mA, ΔI_L = -400mA	V_{OUT}	E_{60}	V	$\frac{\Delta V_{OUT}}{\Delta I_L} = \frac{E_{60}}{200}$	See figure 17	-	0.30	mV/mA

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TABLE III. Group A inspection for device type 05.

Subgroup	Symbol	Test no.	See figure 13 Test conditions V_{IN} (volts)	I_L (mA)	Applied test voltages (volts) (Hi-Lo pin potential) 1-2	4-5	6-11	7-2	8-2	Relays energized	Measurement sense lines Pins	Value	Unit	Equation	Notes	Limits Min	Max	Unit
1 (T_C = 25°C)	V_{OUT1}	1	4.25	-5	15	0	-	-0.425	-	None	9-11	E_1	V	$V_{OUT1} = E_1$		1.20	1.30	V
	V_{OUT2}	2	4.25	-3000	15	-2.995	-	-0.425	-			E_2		$V_{OUT2} = E_2$				
	V_{OUT3}	3	36.25	-5	42.5	0	-	-3.625	-			E_3		$V_{OUT3} = E_3$				
	V_{OUT4}	4	36.25	-150	42.5	-0.145	-	-3.625	-			E_4		$V_{OUT4} = E_4$				
	V_{RLINE}	5	-	-	-	-	-	-	-			-		$V_{RLINE} = E_1 - E_3$	See figure 13 waveforms	-4	4	mV
	V_{RLOAD1}	6	6.25	-5	15	0	-	-0.625	-			E_5			See figure 13 waveforms	-3.5	3.5	
	V_{RLOAD1}	7	6.25	-3000	15	-2.995	-	-0.625	-			E_6		$V_{RLOAD1} = E_5 - E_6$				
	V_{RLOAD2}	8	-	-	-	-	-	-	-			-		$V_{RLOAD2} = E_3 - E_4$		-3.5	3.5	
	V_{RTH}	9	11.25	-1000	25	-0.995	-	-1.125	-			E_7		$V_{RTH} = E_7$	See figure 13 waveforms t = 20.5ms	-5	5	
	I_{ADJ}	10	4.25	-5	15	0	-	-0.425	-	K2	12-13	E_8	mV	$I_{ADJ} = \frac{E_8}{2000}$		-100	-15	µA
	I_{ADJ}	11	36.25	-5	42.5	0	-	-3.625	-			E_9		$I_{ADJ} = \frac{E_9}{2000}$		-100	-15	
	ΔI_{ADJ} (LINE)	12	-	-	-	-	-	-	-			-		ΔI_{ADJ} (LINE) $= \frac{E_8 - E_9}{2000}$		-5	5	
	ΔI_{ADJ} (LOAD)	13	6.25	-5	15	0	-	-0.625	-			E_{10}		ΔI_{ADJ} (LOAD) $= \frac{E_{10} - E_{11}}{2000}$		-5	5	
	ΔI_{ADJ} (LOAD)	14	6.25	-3000	15	-2.995	-	-0.625	-			E_{11}						
	I_{OS1}	15	4.25	-	15	-	-	-0.425	0	K4,K5	10-5	E_{12}	V	$I_{OS1} = 4E_{12}$	See figure 13 waveforms	-5.2	-3.0	A
	V_{OUT5} (RECOV)	16	4.25	-	15	-	-	-0.425	15	K4,K5	9-11	E_{13}		$V_{OUT5} = E_{13}$ (RECOV)		1.20	1.30	V
	I_{OS2}	17	35	-	42.5	-	-	-3.5	0	K5	10-5	E_{14}		$I_{OS2} = 4E_{14}$		-2.0	-0.15	A
	V_{OUT6} (RECOV)	18	35	-	42.5	-	-	-3.5	15	K5	9-11	E_{15}		$V_{OUT6} = E_{15}$ (RECOV)		1.20	1.30	V
	I_{Q1}	19	4.25	-	15	0	1.4	-0.425	-	K3	12-13	E_{16}		$I_{Q1} = \frac{E_{16}}{2000}$		-3.0	-0.5	mA
	I_{Q2}	20	14.25	-	25	0	1.4	-1.425	-	K3	12-13	E_{17}		$I_{Q2} = \frac{E_{17}}{2000}$		-3.0	-0.5	mA
	I_{Q3}	21	36.25	-	42.5	0	1.4	-3.625	-	K3	12-13	E_{18}		$I_{Q3} = \frac{E_{18}}{2000}$		-5.0	-0.5	mA
	V_{START}	22	4.25	-3000	15	0	-	-0.425	-	K4	9-11	E_{19}		$V_{START} = E_{19}$	See figure 13 waveforms	1.20	1.30	V
2 (T_C = 125°C)	V_{OUT1}	23	4.25	-5	15	0	-	-0.425	-	None		E_{20}		$V_{OUT1} = E_{20}$				
	V_{OUT2}	24	4.25	-3000	15	-2.995	-	-0.425	-			E_{21}		$V_{OUT2} = E_{21}$				
	V_{OUT3}	25	36.25	-5	42.5	0	-	-3.625	-			E_{22}		$V_{OUT3} = E_{22}$				
	V_{OUT4}	26	36.25	-150	42.5	-0.145	-	-3.625	-			E_{23}		$V_{OUT4} = E_{23}$				
	V_{RLINE}	27	-	-	-	-	-	-	-			-		$V_{RLINE} = E_{20} - E_{22}$	See figure 13 waveforms	-20	20	mV
	V_{RLOAD1}	28	6.25	-5	15	0	-	-0.625	-			E_{24}			See figure 13 waveforms			
	V_{RLOAD1}	29	6.25	-3000	15	-2.995	-	-0.625	-			E_{25}		$V_{RLOAD1} = E_{24} - E_{25}$		-12	12	
	V_{RLOAD2}	30	-	-	-	-	-	-	-			-		$V_{RLOAD2} = E_{22} - E_{23}$		-12	12	

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TABLE III. Group A inspection for device type 05 - Continued.

Subgroup	Symbol	Test no.	See figure 13 Test conditions V_{IN} (volts)	I_L (mA)	Applied test voltages (volts) (Hi-Lo pin potential) 1-2	4-5	6-11	7-2	8-2	Relays energized	Measurement sense lines Pins	Value	Unit	Equation	Notes	Limits Min	Max	Unit
2 (T_C = 125°C)	I_{ADJ}	31	4.25	-5	15	0	-	-0.425	-	K2	12-13	E_{26}	mV	$I_{ADJ} = \frac{E_{26}}{2000}$		-100	-15	µA
	I_{ADJ}	32	36.25	-5	42.5	0	-	-3.625	-			E_{27}		$I_{ADJ} = \frac{E_{27}}{2000}$		-100	-15	
	ΔI_{ADJ} (LINE)	33	-	-	-	-	-	-	-			-		ΔI_{ADJ} (LINE) $= \frac{E_{26}-E_{27}}{2000}$		-5	5	
	ΔI_{ADJ} (LOAD)	34	6.25	-5	15	0	-	-0.625	-			E_{28}		ΔI_{ADJ} (LOAD) $= \frac{E_{28}-E_{29}}{2000}$		-5	5	
	ΔI_{ADJ} (LOAD)	35	6.25	-3000	15	-2.995	-	-0.625	-			E_{29}						
	I_{OS1}	36	4.25	-	15	-	-	-0.425	0	K4,K5	10-5	E_{30}	V	$I_{OS1} = 4E_{30}$	See figure 13 waveforms	-5.2	-3.0	A
	V_{OUT5} (RECOV)	37	4.25	-	15	-	-	-0.425	15	K4,K5	9-11	E_{31}		$V_{OUT5} = E_{31}$ (RECOV)		1.20	1.30	V
	I_{OS2}	38	35	-	42.5	-	-	-3.5	0	K5	10-5	E_{32}		$I_{OS2} = 4E_{32}$		-2.0	-0.15	A
	V_{OUT6} (RECOV)	39	35	-	42.5	-	-	-3.5	15	K5	9-11	E_{33}		$V_{OUT6} = E_{33}$ (RECOV)		1.20	1.30	V
	I_{Q1}	40	4.25	-	15	0	1.4	-0.425	-	K3	12-13	E_{34}		$I_{Q1} = \frac{E_{34}}{2000}$		-3.0	-0.5	mA
	I_{Q2}	41	14.25	-	25	0	1.4	-1.425	-	K3	12-13	E_{35}		$I_{Q2} = \frac{E_{35}}{2000}$		-3.0	-0.5	
	I_{Q3}	42	36.25	-	42.5	0	1.4	-3.625	-	K3	12-13	E_{36}		$I_{Q3} = \frac{E_{36}}{2000}$		-5.0	-0.5	
	V_{START}	43	4.25	-3000	15	0	-	-0.425	-	K4	9-11	E_{37}		$V_{START} = E_{37}$	See figure 13 waveforms	1.20	1.30	V
(T_C = 150°C)	V_{OUT7}	44	6.25	-5	15	0	-	-0.625	-	None		E_{38}		$V_{OUT7} = E_{38}$				
3 (T_C = -55°C)	V_{OUT1}	45	4.25	-5	15	0	-	-0.425	-			E_{39}		$V_{OUT1} = E_{39}$				
	V_{OUT2}	46	4.25	-3000	15	-2.995	-	-0.425	-			E_{40}		$V_{OUT2} = E_{40}$				
	V_{OUT3}	47	36.25	-5	42.5	0	-	-3.625	-			E_{41}		$V_{OUT3} = E_{41}$				
	V_{OUT4}	48	36.25	-150	42.5	-0.145	-	-3.625	-			E_{42}		$V_{OUT4} = E_{42}$				
	V_{RLINE}	49	-	-	-	-	-	-	-			-		$V_{RLINE} = E_{39}-E_{41}$	See figure 13 waveforms	-20	20	mV
	V_{RLOAD1}	50	6.25	-5	15	0	-	-0.625	-			E_{43}			See figure 13 waveforms			
	V_{RLOAD1}	51	6.25	-3000	15	-2.995	-	-0.625	-			E_{44}		$V_{RLOAD1} = E_{43}-E_{44}$		-12	12	
	V_{RLOAD2}	52	-	-	-	-	-	-	-			-		$V_{RLOAD2} = E_{41}-E_{42}$		-12	12	
	I_{ADJ}	53	4.25	-5	15	0	-	-0.425	-	K2	12-13	E_{45}	mV	$I_{ADJ} = \frac{E_{45}}{2000}$		-100	-15	µA
	I_{ADJ}	54	36.25	-5	42.5	0	-	-3.625	-			E_{46}		$I_{ADJ} = \frac{E_{46}}{2000}$		-100	-15	
	ΔI_{ADJ} (LINE)	55	-	-	-	-	-	-	-			-		ΔI_{ADJ} (LINE) $= \frac{E_{45}-E_{46}}{2000}$		-5	5	
	ΔI_{ADJ} (LOAD)	56	6.25	-5	15	0	-	-0.625	-			E_{47}		ΔI_{ADJ} (LOAD) $= \frac{E_{47}-E_{48}}{2000}$		-5	5	
	ΔI_{ADJ} (LOAD)	57	6.25	-3000	15	-2.995	-	-0.625	-			E_{48}						

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TABLE III. Group A inspection for device type 05 - Continued.

Subgroup	Symbol	Test no.	See figure 13 Test conditions V_{IN} (volts)	I_L (mA)	Applied test voltages (volts) (Hi-Lo pin potential) 1-2	4-5	6-11	7-2	8-2	Relays energized	Measurement sense lines Pins	Value	Unit	Equation	Notes	Limits Min	Max	Unit
3 (T_C = -55°C)	I_{OS1}	58	4.25	-	15	-	-	-0.425	0	K4,K5	10-5	E_{49}	V	$I_{OS1} = 4E_{49}$	See figure 13 waveforms	-5.2	-3.0	A
	V_{OUT5} (RECOV)	59	4.25	-	15	-	-	-0.425	15	K4,K5	9-11	E_{50}		$V_{OUT5} = E_{50}$ (RECOV)		1.20	1.30	V
	I_{OS2}	60	35	-	42.5	-	-	-3.5	0	K5	10-5	E_{51}		$I_{OS2} = 4E_{51}$		-2.0	-0.15	A
	V_{OUT6} (RECOV)	61	35	-	42.5	-	-	-3.5	15	K5	9-11	E_{52}		$V_{OUT6} = E_{52}$ (RECOV)		1.20	1.30	V
	I_{Q1}	62	4.25	-	15	0	1.4	-0.425	-	K3	12-13	E_{53}		$I_{Q1} = \frac{E_{53}}{2000}$		-3.0	-0.5	mA
	I_{Q2}	63	14.25	-	25	0	1.4	-1.425	-	K3	12-13	E_{54}		$I_{Q2} = \frac{E_{54}}{2000}$		-3.0	-0.5	mA
	I_{Q3}	64	36.25	-	42.5	0	1.4	-3.625	-	K3	12-13	E_{55}		$I_{Q3} = \frac{E_{55}}{2000}$		-5.0	-0.5	mA
	V_{START}	65	4.25	-3000		0	-	-0.425	-	K4	9-11	E_{56}		$V_{START} = E_{56}$	See figure 13 waveforms	1.20	1.30	V

Subgroup	Symbol	Test no.	Test conditions Input voltage	Load current	Measurement sense lines Symbol	Value	Unit	Equation	Notes	Limits Min	Max	Unit
4 (T_C = 25°C)	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	66	V_{IN} = 6.25V, e_i = 1.0Vrms @ 2400 Hz	I_L = -500mA	e_{orms}	E_{57}	Vrms	$\frac{\Delta V_{IN}}{\Delta V_{OUT}} = -20\log E_{57}$	See figure 14	65	-	dB
7 (T_C = 25°C)	V_{NO}	67	V_{IN} = 6.25V	I_L = -100mA	e_{orms}	E_{58}	Vrms	$V_{NO} = E_{58}$	See figure 15	-	120	µVrms
	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	68	V_{IN} = 6.25V, ΔV_{IN} = 3.0V	I_L = -10mA	V_{OUT}	E_{59}	V	$\frac{\Delta V_{OUT}}{\Delta V_{IN}} = \frac{E_{59}}{3}$	See figure 16	-	12	mV/V
	$\frac{\Delta V_{OUT}}{\Delta I_L}$	69	V_{IN} = 6.25V	I_L = -100mA, ΔI_L = -400mA	V_{OUT}	E_{60}	V	$\frac{\Delta V_{OUT}}{\Delta I_L} = \frac{E_{60}}{200}$	See figure 17	-	0.30	mV/mA

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TABLE III. Group A inspection for device type 06.

Subgroup	Symbol	Test no.	See figure 13 Test conditions V_{IN} (volts)	I_L (mA)	Applied test voltages (volts) (Hi-Lo pin potential) 1-2	4-5	6-11	7-2	8-2	Relays energized	Measurement sense lines Pins	Value	Unit	Equation	Notes	Limits Min	Max	Unit
1 (T_C = 25°C)	V_{OUT1}	1	4.25	-5	15	0	-	-0.425	-	None	9-11	E_1	V	$V_{OUT1} = E_1$		1.19	1.29	V
	V_{OUT2}	2	4.25	-5000	15	-4.995	-	-0.425	-			E_2		$V_{OUT2} = E_2$				
	V_{OUT3}	3	36.25	-5	42.5	0	-	-3.625	-			E_3		$V_{OUT3} = E_3$				
	V_{OUT4}	4	36.25	-0.150	42.5	-0.145	-	-3.625	-			E_4		$V_{OUT4} = E_4$				
	V_{OUT5}	5	6.25	-7000	15	-6.995	-	-0.625	-			E_5		$V_{OUT5} = E_5$				
	V_{RLINE}	6	-	-	-	-	-	-	-			-		$V_{RLINE} = E_1 - E_3$	See figure 13 waveforms	-4	4	mV
	V_{RLOAD1}	7	6.25	-5	15	0	-	-0.625	-			E_6			See figure 13 waveforms	-3.8	3.8	
	V_{RLOAD1}	8	6.25	-5000	15	-4.995	-	-0.625	-			E_7		$V_{RLOAD1} = E_6 - E_7$				
	V_{RLOAD2}	9	-	-	-	-	-	-	-			-		$V_{RLOAD2} = E_3 - E_4$		-3.8	3.8	
	V_{RTH}	10	11.25	-1000	25	-0.995	-	-1.125	-			E_8		$V_{RTH} = E_8$	See figure 13 waveforms t = 20.5ms	-2	2	
	I_{ADJ}	11	4.25	-5	15	0	-	-0.425	-	K2	12-13	E_9	mV	$I_{ADJ} = \frac{E_9}{2000}$		-100	-15	µA
	I_{ADJ}	12	36.25	-5	42.5	0	-	-3.625	-			E_{10}		$I_{ADJ} = \frac{E_{10}}{2000}$		-100	-15	
	ΔI_{ADJ} (LINE)	13	-	-	-	-	-	-	-			-		ΔI_{ADJ} (LINE) $= \frac{E_9 - E_{10}}{2000}$		-5	5	
	ΔI_{ADJ} (LOAD)	14	6.25	-5	15	0	-	-0.625	-			E_{11}		ΔI_{ADJ} (LOAD) $= \frac{E_{11} - E_{12}}{2000}$		-5	5	
	ΔI_{ADJ} (LOAD)	15	6.25	-5000	15	-4.995	-	-0.625	-			E_{12}						
	I_{OS1}	16	4.25	-	15	-	-	-0.425	0	K4,K5	10-5	E_{13}	V	$I_{OS1} = 4E_{13}$	t = 0.1 ms	-16.0	-7.0	A
	I_{OS2}	17	4.25	-	15	-	-	-0.425	0	K4,K5		E_{14}		$I_{OS2} = 4E_{14}$	t = 0.5 ms	-16.0	-7.0	A
	I_{OS3}	18	4.25	-	15	-	-	-0.425	0	K4,K5		E_{15}		$I_{OS3} = 4E_{15}$	t = 5.0 ms	-15.0	-5.0	A
	V_{OUT6} (RECOV)	19	4.25	-	15	-	-	-0.425	15	K4,K5	9-11	E_{16}		V_{OUT6} (RECOV) $= E_{16}$		1.19	1.29	V
	I_{OS4}	20	35	-	42.5	-	-	-3.5	0	K5	10-5	E_{17}		$I_{OS4} = 4E_{17}$	See figure 13 waveforms	-3.0	-0.20	A
	V_{OUT7} (RECOV)	21	35	-	42.5	-	-	-3.5	15	K5	9-11	E_{18}		V_{OUT7} (RECOV) $= E_{18}$		1.19	1.29	V
	I_{Q1}	22	4.25	-	15	0	1.4	-0.425	-	K3	12-13	E_{19}		$I_{Q1} = \frac{E_{19}}{2000}$		-3.0	-0.5	mA
	I_{Q2}	23	14.25	-	25	0	1.4	-1.425	-	K3	12-13	E_{20}		$I_{Q2} = \frac{E_{20}}{2000}$		-3.0	-0.5	mA
	I_{Q3}	24	36.25	-	42.5	0	1.4	-3.625	-	K3	12-13	E_{21}		$I_{Q3} = \frac{E_{21}}{2000}$		-5.0	-0.50	mA
	V_{START}	25	4.25	-5000	15	0	-	-0.425	-	K4	9-11	E_{22}		$V_{START} = E_{22}$	See figure 13 waveforms	1.19	1.29	V
2 (T_C = 125°C)	V_{OUT1}	26	4.25	-5	15	0	-	-0.425	-	None		E_{23}		$V_{OUT1} = E_{23}$				
	V_{OUT2}	27	4.25	-5000	15	-4.995	-	-0.425	-			E_{24}		$V_{OUT2} = E_{24}$				
	V_{OUT3}	28	36.25	-5	42.5	0	-	-3.625	-			E_{25}		$V_{OUT3} = E_{25}$				
	V_{OUT4}	29	36.25	-0.150	42.5	-0.145	-	-3.625	-			E_{26}		$V_{OUT4} = E_{26}$				
	V_{OUT5}	30	6.25	-7000	15	-6.995	-	-0.625	-			E_{27}		$V_{OUT5} = E_{27}$				
	V_{RLINE}	31	-	-	-	-	-	-	-			E_{28}		$V_{RLINE} = E_{23} - E_{25}$	See figure 13 waveforms	-20	20	mV
	V_{RLOAD1}	32	6.25	-5	15	0	-	-0.625	-			E_{29}			See figure 13 waveforms	-8	8	
	V_{RLOAD1}	33	6.25	-5000	15	-4.995	-	-0.625	-			E_{30}		$V_{RLOAD1} = E_{29} - E_{30}$		-8	8	
	V_{RLOAD2}	34	-	-	-	-	-	-	-			-		$V_{RLOAD2} = E_{25} - E_{26}$				

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TABLE III. Group A inspection for device type 06 - Continued.

Subgroup	Symbol	Test no.	See figure 13: Test conditions V_{IN} (volts)	I_L (mA)	Applied test voltages (volts) (Hi-Lo pin potential) 1-2	4-5	6-11	7-2	8-2	Relays energized	Measurement sense lines: Pins	Value	Unit	Equation	Notes	Limits: Min	Max	Unit
2 (T_C = 125°C)	I_{ADJ}	35	4.25	-5	15	0	-	-0.425	-	K2	12-13	E_{31}	mV	$I_{ADJ} = \frac{E_{31}}{2000}$		-100	-15	µA
	I_{ADJ}	36	36.25	-5	42.5	0	-	-3.625	-			E_{32}		$I_{ADJ} = \frac{E_{32}}{2000}$		-100	-15	
	ΔI_{ADJ} (LINE)	37	-	-	-	-	-	-	-			-		ΔI_{ADJ} (LINE) $= \frac{E_{31}-E_{32}}{2000}$		-5	5	
	ΔI_{ADJ} (LOAD)	38	6.25	-5	15	0	-	-0.625	-			E_{33}		ΔI_{ADJ} (LOAD) $= \frac{E_{33}-E_{34}}{2000}$		-5	5	
	ΔI_{ADJ} (LOAD)	39	6.25	-5000	15	-4.995	-	-0.625	-			E_{34}						
	I_{OS1}	40	4.25	-	15	-	-	-0.425	0	K4,K5	10-5	E_{35}	V	$I_{OS1} = 4E_{35}$	t = 0.1 ms	-16.0	-7.0	A
	I_{OS2}	41	4.25	-	15	-	-	-0.425	0	K4,K5	10-5	E_{36}		$I_{OS2} = 4E_{36}$	t = 0.5 ms	-16.0	-7.0	A
	I_{OS3}	42	4.25	-	15	-	-	-0.425	0	K4,K5	10-5	E_{37}		$I_{OS3} = 4E_{37}$	t = 5.0 ms	-15.0	-5.0	A
	V_{OUT6} (RECOV)	43	4.25	-	15	-	-	-0.425	15	K4,K5	9-11	E_{38}		$V_{OUT6} = E_{38}$ (RECOV)		1.19	1.29	V
	I_{OS4}	44	35	-	42.5	-	-	-3.5	0	K5	10-5	E_{39}		$I_{OS4} = 4E_{39}$	t = 10.0 ms	-3.0	-0.20	A
	V_{OUT7} (RECOV)	45	35	-	42.5	-	-	-3.5	15	K5	9-11	E_{40}		$V_{OUT6} = E_{40}$ (RECOV)	See figure 13 waveforms	1.19	1.29	V
	I_{Q1}	46	4.25	-	15	0	1.4	-0.425	-	K3	12-13	E_{41}		$I_{Q1} = \frac{E_{41}}{2000}$		-3.0	-0.5	mA
	I_{Q2}	47	14.25	-	25	0	1.4	-1.425	-	K3	12-13	E_{42}		$I_{Q2} = \frac{E_{42}}{2000}$		-3.0	-0.5	
	I_{Q3}	48	36.25	-	42.5	0	1.4	-3.625	-	K3	12-13	E_{43}		$I_{Q3} = \frac{E_{43}}{2000}$		-5.0	-0.5	
	V_{START}	49	4.25	-5000	15	0	-	-0.425	-	K4	9-11	E_{44}		$V_{START} = E_{44}$	See figure 13 waveforms	1.19	1.29	V
(T_C = 150°C)	V_{OUT7}	50	6.25	-5	15	0	-	-0.625	-	None		E_{45}		$V_{OUT7} = E_{45}$				
3 (T_C = -55°C)	V_{OUT1}	51	4.25	-5	15	0	-	-0.425	-			E_{46}		$V_{OUT1} = E_{46}$				
	V_{OUT2}	52	4.25	-5000	15	-4.995	-	-0.425	-			E_{47}		$V_{OUT2} = E_{47}$				
	V_{OUT3}	53	36.25	-5	42.5	0	-	-3.625	-			E_{48}		$V_{OUT3} = E_{48}$				
	V_{OUT4}	54	36.25	-0.150	42.5	-0.145	-	-3.625	-			E_{49}		$V_{OUT4} = E_{49}$				
	V_{OUT5}	55	6.25	-7000	15	-6.995	-	-0.625	-			E_{50}		$V_{OUT5} = E_{50}$				
	V_{RLINE}	56	-	-	-	-	-	-	-			-		$V_{RLINE} = E_{46}-E_{48}$	See figure 13 waveforms	-17	17	mV
	V_{RLOAD1}	57	6.25	-5	15	0	-	-0.625	-			E_{51}			See figure 13 waveforms			
	V_{RLOAD1}	58	6.25	-5000	15	-4.995	-	-0.625	-			E_{52}		$V_{RLOAD1} = E_{51}-E_{52}$		-8	8	
	V_{RLOAD2}	59	-	-	-	-	-	-	-			-		$V_{RLOAD2} = E_{48}-E_{49}$		-8	8	
	I_{ADJ}	60	4.25	-5	15	0	-	-0.425	-	K2	12-13	E_{53}	mV	$I_{ADJ} = \frac{E_{53}}{2000}$		-100	-15	µA
	I_{ADJ}	61	36.25	-5	42.5	0	-	-3.625	-			E_{54}		$I_{ADJ} = \frac{E_{54}}{2000}$		-100	-15	
	ΔI_{ADJ} (LINE)	62	-	-	-	-	-	-	-			-		ΔI_{ADJ} (LINE) $= \frac{E_{53}-E_{54}}{2000}$		-5	5	
	ΔI_{ADJ} (LOAD)	63	6.25	-5	15	0	-	-0.625	-			E_{55}		ΔI_{ADJ} (LOAD) $= \frac{E_{55}-E_{56}}{2000}$		-5	5	
	ΔI_{ADJ} (LOAD)	64	6.25	-5000	15	-4.995	-	-0.625	-			E_{56}						

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TABLE III. Group A inspection for device type 06 - Continued.

Subgroup	Symbol	Test no.	See figure 13. Test conditions V_{IN} (volts)	Test conditions I_L (mA)	Applied test voltages (volts) (Hi-Lo pin potential) 1-2	4-5	6-11	7-2	8-2	Relays energized	Measurement sense lines Pins	Value	Unit	Equation	Notes	Limits Min	Limits Max	Unit
3 T_C = -55°C	I_{OS1}	65	4.25	-	15	-	-	-0.425	0	K4,K5	10-5	E_{57}	V	$I_{OS1} = 4E_{57}$	t = 0.1 ms	-16.0	-7.0	A
	I_{OS2}	66	4.25	-	15	-	-	-0.425	0	K4,K5		E_{58}		$I_{OS2} = 4E_{58}$	t = 0.5 ms	-16.0	-7.0	A
	I_{OS3}	67	4.25	-	15	-	-	-0.425	0	K4,K5		E_{59}		$I_{OS3} = 4E_{59}$	t = 5.0 ms	-15.0	-5.0	A
	V_{OUT6} (RECOV)	68	4.25	-	15	-	-	-0.425	15	K4,K5	9-11	E_{60}		$V_{OUT6} = E_{60}$ (RECOV)		1.19	1.29	V
	I_{OS2}	69	35	-	42.5	-	-	-3.5	0	K5	10-5	E_{61}		$I_{OS4} = 4E_{61}$	t = 10.0 ms	-3.0	-0.20	A
	V_{OUT7} (RECOV)	70	35	-	42.5	-	-	-3.5	15	K5	9-11	E_{62}		$V_{OUT7} = E_{62}$ (RECOV)	See figure 13 waveforms	1.19	1.29	V
	I_{Q1}	71	4.25	-	15	0	1.4	-0.425	-	K3	12-13	E_{63}		$I_{Q1} = \frac{E_{63}}{2000}$		-3.0	-0.5	mA
	I_{Q2}	72	14.25	-	25	0	1.4	-1.425	-	K3	12-13	E_{64}		$I_{Q2} = \frac{E_{64}}{2000}$		-3.0	-0.5	mA
	I_{Q3}	73	36.25	-	42.5	0	1.4	-3.625	-	K3	12-13	E_{65}		$I_{Q3} = \frac{E_{65}}{2000}$		-5.0	-0.5	mA
	V_{START}	74	4.25	-5000	15	0	-	-0.425	-	K4	9-11	E_{66}		$V_{START} = E_{66}$	See figure 13 waveforms	1.19	1.29	V

Subgroup	Symbol	Test no.	Test conditions: Input voltage	Test conditions: Load current	Measurement sense lines Symbol	Value	Unit	Equation	Notes	Limits Min	Limits Max	Unit
4 (T_C = 25°C)	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	75	V_{IN} = 6.25V, e_i = 1.0Vrms @ 2400 Hz	I_L = -500mA	e_{orms}	E_{67}	Vrms	$\frac{\Delta V_{IN}}{\Delta V_{OUT}} = -20\log E_{67}$	See figure 14	65	-	dB
7 (T_C = 25°C)	V_{NO}	76	V_{IN} = 6.25V	I_L = -100mA	e_{orms}	E_{68}	Vrms	$V_{NO} = E_{68}$	See figure 15	-	120	μVrms
	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	77	V_{IN} = 6.25V, ΔV_{IN} = 3.0V	I_L = -10mA	V_{OUT}	E_{69}	V	$\frac{\Delta V_{OUT}}{\Delta V_{IN}} = \frac{E_{69}}{3}$	See figure 16	-	12	mV/V
	$\frac{\Delta V_{OUT}}{\Delta I_L}$	78	V_{IN} = 6.25V	I_L = -100mA, ΔI_L = -400mA	V_{OUT}	E_{70}	V	$\frac{\Delta V_{OUT}}{\Delta I_L} = \frac{E_{70}}{200}$	See figure 17	-	0.3	mV/mA

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4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Subgroups 5, 6, 8, 9, 10, and 11 shall be omitted.

b. Tests shall be as specified in table II herein.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Life test for class S shall be performed at 125°C for 1000 hours at an LTPD of 5 (per group C, subgroup 2) as a substitute for subgroup 5 of table IIa. If the sample fails the LTPD acceptance, individual devices in the lot may be acceptable if they pass 1000 hours additional burn-in and the lot PDA does not exceed 5 percent.

c. Constant acceleration for class S (method 2001 of MIL-STD-883); test condition B shall be used for case Y.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup I of group C inspection for classes B and C devices.

b. Life tests for class B and C (method 1005 of MIL-STD-883), test condition C only using the circuits shown on figure 11. Accelerated test conditions shall not be used.

c. Constant acceleration (method 2001 of MIL-STD-883); test condition b shall be used for case Y.

4.4.4 Group D inpection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Constant acceleration (method 2001 of MIL-STD-883); test condition B shall be used for case Y.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables. Electrical test circuits as prescribed herein or in the referenced test methods of MIL-STD-883 shall be acceptable. Other test circuits shall require the approval of the qualifying activity.

4.5.1 Voltage and current. All voltage values given are referenced to the designated return sense line. Currents given are conventional current and positive when flowing into the referenced terminal.

4.5.2 Life test cooldown procedure. When devices are measured at 25°C following application of the operating life or burn-in test condition, they shall be cooled to room temperature prior to removal of the bias.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

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TABLE IV. Group C end-point electrical parameters.

Device type	Characteristic	Symbol	Delta limits 1/	Limits Min	Limits Max	Unit
01,02	Output voltage	V_{OUT}	±50 mV	4.75	5.25	V
01,02	Standby current drain	I_{SCD}	±20%	-8.0	-0.5	mA
03,04,05	Output voltage	V_{OUT}	±10 mV	1.20	1.30	V
06	Output voltage	V_{OUT}	±10 mV	1.19	1.29	V
03,04,05,06	Adjust pin current	I_{ADJ}	±10 µA	-100	-15	µA
03,04	Line regulation	V_{RLINE}	±4 mV	-9	9	mV
05,06	Line regulation	V_{RLINE}	±2 mV	-4	4	mV

1/ Delta limits apply to the measured value (see delta limit definition in MIL-M-38510).

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.3 Ordering data. The contract or purchase order should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to procuring activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for carriers, special lead lengths or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

6.4 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

6.4.1 Line regulation. The change in output voltage for a specified change in input voltage (V_{RLINE}).

6.4.2 Load regulation. The change in output voltage for a specified change in load current (V_{RLOAD}).

6.4.3 Ripple rejection. The ratio of the peak-to-peak input ripple voltage to the peak-to-peak output ripple voltage ($\Delta V_{OUT}/\Delta V_{IN}$).

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6.4.4 Output noise voltage. The rms output noise voltage with constant load and no input ripple (V_{NO}).

6.4.5 Standby current drain. The supply current drawn by the regulator with no output load or with a 1 kilohm output load.

6.4.6 Minimum load current. The minimum load current is that current required to maintain regulation.

6.4.7 Input voltage range. The range of supply voltage over which the regulator will operate.

6.4.8 Output voltage range. The range of output voltage over which the regulator will operate.

6.4.9 Transient response. The closed-loop step function response of the regulator under small-signal conditions.

6.5 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits procured for Government logistic support will be procured to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	78MG
02	78G
03	LM117H
04	LM117K
05	LM150K
06	LM138K

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - MI, AR
Navy - SH
Air Force - 11, 19, 85, 99
DLA - ES
NASA - NA

User activities:
Army - SM
Navy - CG, MC, AS, OS

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-M-38510/117A, Microcircuits, Linear, Adjustable, Positive, Voltage Regulators, Monolithic Silicon

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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